

BASIC LAB SKILLS



Kari Christenson
Western Technical College

Basic Lab Skills

by Kari Christenson

Western
Technical College

2025

This text is disseminated via the Open Education Resource (OER) LibreTexts Project (<https://LibreTexts.org>) and like the thousands of other texts available within this powerful platform, it is freely available for reading, printing, and "consuming."

The LibreTexts mission is to bring together students, faculty, and scholars in a collaborative effort to provide an accessible, and comprehensive platform that empowers our community to develop, curate, adapt, and adopt openly licensed resources and technologies; through these efforts we can reduce the financial burden born from traditional educational resource costs, ensuring education is more accessible for students and communities worldwide.

Most, but not all, pages in the library have licenses that may allow individuals to make changes, save, and print this book. Carefully consult the applicable license(s) before pursuing such effects. Instructors can adopt existing LibreTexts texts or Remix them to quickly build course-specific resources to meet the needs of their students. Unlike traditional textbooks, LibreTexts' web based origins allow powerful integration of advanced features and new technologies to support learning.



LibreTexts is the adaptable, user-friendly non-profit open education resource platform that educators trust for creating, customizing, and sharing accessible, interactive textbooks, adaptive homework, and ancillary materials. We collaborate with individuals and organizations to champion open education initiatives, support institutional publishing programs, drive curriculum development projects, and more.

The LibreTexts libraries are Powered by [NICE CXone Expert](#) and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation nor the US Department of Education.

Have questions or comments? For information about adoptions or adaptations contact info@LibreTexts.org or visit our main website at <https://LibreTexts.org>.

The LibreTexts name, logo, and book covers are trademarked by LibreTexts, Inc. (a 501(c)(3) not-for-profit organization) and protected against unauthorized use.

This text was compiled on 08/08/2026

- [InfoPage](#) by [Delmar Larsen](#) is licensed [Public Domain](#).

Introduction

Welcome to Basic Lab Skills!

This resource provides the text for Basic Lab Skills, a first year, first term course in the Medical Laboratory Technician Program at Western Technical College. This course covers the basic skills needed in a laboratory environment.

Students will explore health careers, how to practice laboratory functions safely and common equipment used in a laboratory environment. A special emphasis will be placed on the proper use and care of a microscope. Key concepts of the Clinical Laboratory Improvement Amendments (CLIA) will be reviewed.

Except where otherwise noted, this text is licensed [Creative Commons-Attribution-NonCommercial-ShareAlike 4.0](https://creativecommons.org/licenses/by-nc-sa/4.0/).

Suggested attribution:

Christenson, K. (2025). *Basic Lab Skills*. Western Technical College. CC BY-NC-SA.

TABLE OF CONTENTS

Introduction

Acknowledgements

Licensing

1: Lab Safety

2: Basic Lab Equipment

3: Microscope

4: Medical Terminology

5: Health Careers

6: CLIA

Index

Glossary

Detailed Licensing

Acknowledgements

This Open Educational Resource was created by Kari Christenson, MLT Program Director/Faculty, with support from an [Open Western OER Fellowship](#) grant provided by Western Technical College and the Wisconsin Technical College System. Project management and editorial support by Ellen Range, Student Learning Librarian. Proofreading and peer review by LeeAnne Schwan, Medical Lab Technician Instructor. Additional peer reviewers included MLT instructors Karen Farner, Southwest Wisconsin Technical College, and Linda Muraski, Northeast Wisconsin Technical College.

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

1: Lab Safety

Learning Objectives

Course Competency:

Adhere to safety and infection control policies

Learning Objectives:

- Explain physical safety policies
- Explain chemical safety policies
- Adhere to biological safety policies
- Practice infection control

Introduction

Laboratory safety involves all the measures taken by the laboratory worker, institution, and regulatory agencies to eliminate potential harm to human health and well-being. Although steps are taken to reduce risks in the workplace, safety is a matter of personal responsibility. A clinical lab may have several safety hazards that must be known and understood by all students or employees working in the lab. It is the responsibility of each person in the lab to know and follow basic laboratory safety rules, to understand how to safely operate equipment, to understand the hazards of materials they are working with and to work to reduce potential risks. In the event of an accident, it is critical to know the location and use of emergency equipment. Having this knowledge should help to prevent accidents and minimize damage that might occur in the event of an accident.

Occupational Safety and Health Administration (OSHA) - It's the Law¹

Creating a safe laboratory workspace is required by law. The federal government regulates safety within the workplace thru the Occupational Safety and Health Administration (OSHA). The regulations are designed to inform workers about hazards they may encounter and protect workers from harm. This is called OSHA's **Hazard Communication Standard**.² Employers must provide safety training, proper safety apparel, and a safe work environment.

Personal Protective Equipment (PPE)³

Employers must provide Personal Protective Equipment (PPE) to protect employees. These items need to protect the face, eyes and skin. Laboratory PPE includes lab coats, face shields, safety glasses and gloves.

For More Information: [This CDC document demonstrates the sequence for putting on \(donning\) and removing \(doffing\) PPE.](#)



Figure 1.1: Laboratory Worker Wearing Personal Protective Equipment. (Copyright DataBase Center for Life Science (DBCLS) via [Wikimedia Commons](#); CC BY 4.0)

Categories of Hazards

Hazards within the laboratory can be divided into **physical hazards**, **chemical hazards**, and **biological hazards**.

Physical Hazards

Physical hazards include electrical and fire safety. Before servicing any equipment, it should be unplugged. Circuits should not be overloaded, and the use of extension cords should be avoided. Fire safety includes storing flammable chemicals in a flameproof cabinet, knowing your facilities escape plan, and the location of fire alarms and extinguishers.

Lab Safety Equipment: Fire Extinguishers⁴

Portable fire extinguishers apply an extinguishing agent that will either cool burning fuel, displace or remove oxygen, or stop the chemical reaction so a fire cannot continue to burn. When the handle of an extinguisher is compressed, agent is expelled out the nozzle.

Most fire extinguishers operate using the following P.A.S.S. technique:

1. **PULL**... Pull the pin. This will also break the tamper seal.
2. **AIM**... Aim low, pointing the extinguisher nozzle (or its horn or hose) at the base of the fire.
3. **SQUEEZE**... Squeeze the handle to release the extinguishing agent.
4. **SWEEP**... Sweep from side to side at the base of the fire until it appears to be out. Watch the area. If the fire re-ignites, repeat steps 2 - 4.

If you have the slightest doubt about your ability to fight a fire, evacuate immediately!

 Portable fire extinguisher

Figure 1.2: Portable Fire Extinguisher. (Copyright [Occupational Safety and Health Administration, U.S. Department of Labor](#); Public Domain)

Chemical Hazards

Chemical hazards are another kind of hazard within the laboratory. OSHA mandates the use of a written safety plan for hazardous chemicals which is called the **Chemical Hygiene Plan**. The Hazard Communication Standard also mandates the use of **Safety Data Sheets (SDS)** and **Hazard Communication Standard (HCS) Labels** to communicate information to the user about chemical hazards.

Safety Data Sheets (SDS)

A Safety Data Sheet (previously called Material Safety Data Sheet) for each chemical is required by OSHA. Sections 1-8 of a SDS contain information about the identification, hazards, composition, safe handling practices, and emergency control measures. Sections 9 through 11 and 16 contain technical and scientific information, such as physical and chemical properties, stability and reactivity, toxicology, and exposure control.

Hazard Communication Standard (HCS) Labels

A quick assessment of a chemical's hazards is visible on the container's Hazard Communication Standard (HCS) label. HCS labeling does not replace the more detailed Safety Data Sheet (SDS) but rather gives the following information in brief:

- Name, Address, and Phone Number of Responsible Party (i.e., the manufacturer or distributor)
- Chemical Identification - chemical name and code or batch number that matches the information found in Section 1 of the chemical's SDS
- Signal Word - "DANGER" for more severe hazards or "WARNING" for less severe hazards
- Hazard Statements - a brief description of the hazard(s)
- Precautionary Statements (optional) - prevention, response, storage, and disposal
- Hazard Category Numbers 1-4 (optional) - "1" for the most severe hazard to "4" for the least severe
- Pictograms

Pictograms

Pictograms are red-bordered, diamond shapes that frame a black graphic on a white background, and these symbols depict the type of hazard(s). HCS labels are required by the Occupation Safety and Health Administration (OSHA) and are standardized, having been adopted from the Globally Harmonized System of Classification and Labeling of Chemicals (GHS) set by the United Nations.

Table 1. HCS Pictogram Guide

Pictogram	Hazard Class	Hazard Type	Example Signal Word & Hazard Statement
	Flammables Self-Reactives Self-Heating Pyrophorics Emits Flammable Gas Organic Peroxides	Physical	DANGER Heating may cause a fire
	Explosives Self-Reactives Organic Peroxides	Physical	WARNING Fire or projection hazard
	Gases Under Pressure	Physical	WARNING Contains gas under pressure; may explode if heated
	Corrosive to Metals Corrosive Skin Corrosion/Burns Eye Damage	Physical Health	DANGER Causes severe skin burns and eye damage
	Oxidizer (gases) Oxidizers (solid or liquid)	Physical Health	WARNING May intensify fire; oxidizer
	Acute Toxicity (fatal or toxic)	Health	DANGER Fatal if swallowed
	Carcinogen Mutagenicity Respiratory Sensitizer Reproductive Toxicity Target Organ Toxicity Aspiration Toxicity	Health	DANGER May cause cancer

Pictogram	Hazard Class	Hazard Type	Example Signal Word & Hazard Statement
	Irritant (skin and eyes) Dermal Sensitizer Acute Toxicity (harmful) Narcotic Effects Respiratory Tract Irritation Hazardous to Ozone Layer (optional)	Health Other	WARNING Causes skin irritation
	(optional label) Environmental Toxicity Aquatic Toxicity	Environment.	WARNING Toxic to aquatic life

National Fire Protection Association (NFPA) Hazard Rating System

The National Fire Protection Association (NFPA) Hazard Rating System was designed for emergency workers such as fire responders to be able to swiftly get basic information about the hazards of a chemical from a simple label. The diamond-shaped label is divided into four color-coded squares, and inside each square is printed a ratings code or number from 0-4, with "0" as the least severe hazard to "4" as the most severe.

Inside the red square, the number indicates the flammability rating, blue indicates the health hazard rating, yellow indicates the chemical's instability, and the white square indicates any special hazards.

NFPA HAZARD RATING SYSTEM GUIDE							
0	Will not typically burn	NFPA Hazard Label			0	Stable even under fire conditions.	
1	Ignites only after considerable preheating				1	Normally stable except under high temperatures	
2	Ignites after moderate preheating or at high ambient temperatures				2	Violent changes at high temperatures and pressures	
3	Ignites at almost any temperature				3	May explode at high temperatures or with shock	
4	Vaporizes at normal temperatures and pressures				4	May explode even at normal temperatures and pressure	
0	No health hazards beyond normal combustible material	BIO Biological Hazard ☢ Radioactive			ACID	Acidic	
1	Exposure can cause significant irritation				ALK	Alkaline	
2	Intense or continued exposure can cause incapacitation or injury				COR	Corrosive	
3	Short exposure can cause serious or permanent injury				OX	Oxidizer	
4	Very short exposure can be fatal				W	Reacts Violently or Explosively with Water	
		POI	Poisonous	RAD			

Chemical Hazards and Waste Disposal

Laboratory waste must be disposed of safely and appropriately. Labs must be aware of the school, state and federal guidelines for waste disposal. Many chemicals can NOT be poured down the sink. Be sure you know which chemicals are hazardous, require special storage and must be placed into properly labeled waste containers kept in fume hoods until sent to the proper hazardous waste disposal. Know which materials are considered biohazards and the proper area or container to place them so they can be autoclaved.

Table 2. Medical and Biohazardous Waste Treatment and Disposal Chart, U. California

Type of Waste	Definition	Required Treatment
Chemical Waste	Any solid, liquid, or gaseous material generated in the laboratory that poses a danger to human health or the environment.	This will vary depending on the chemical. The institution you are in will have specific requirements to meet regulatory code. The aim is to reduce hazards and minimize environmental impact. Follow your instructor's directions.

Type of Waste	Definition	Required Treatment
Solid Biohazardous Waste	Materials such as pipettes, petri dishes, or other culture flasks disposable or glass that do not contain liquid but were in contact with cultures of cells or human or animal-derived materials.	Disposable waste is placed in red biohazard bags. Treat and replace bags when they are halfway full. Reusable glass materials must be placed in autoclavable trays. To treat waste, the autoclave must be properly loaded and set for a minimum temperature of 121°C (250°F) for 60 minutes at 15 psi. Autoclave tape must be placed on the bag or tray to indicate it has passed through the cycle. Material should be autoclaved as soon as possible, but at a maximum of 7 days after it is generated.
Liquid Biohazardous Waste	Broth cultures or cell culture media or contaminated liquid media.	Decontaminate using 10% bleach solution for a minimum 2 hours contact time. Dispose down the drain with water. Alternatively, racks of used culture tubes can be autoclaved using the same temperature and time indicated above but set for a slow exhaust liquid setting. Use autoclave tape as above. Treat waste as soon as possible but not longer than 7 days post generation.
Sharps Waste	Any object that is capable of piercing or damaging human skin that is contaminated with chemical, or biohazardous waste e.g., scalpels, blades, needles, broken glass, etc.	Place in the marked sharps container. Remember that broken glass that is not contaminated will have a separate disposal container within the laboratory.

Biological Hazards⁵

Biological hazards are the third kind of hazards within the laboratory environment. All human blood and other body fluids are treated as if they are infectious. This is applied universally to all patients, regardless of whether they are suspected of being infectious. OSHA's **Bloodborne Pathogens Standard**⁶ outlines how to protect workers from bloodborne pathogens or other potentially infectious material (OPIM). Bloodborne pathogens are disease-causing microorganisms in human blood or OPIM. The standard requires the employer to train lab staff members who might be exposed to blood and/or OPIM.

Exposure is defined as contact with blood or OPIM at the eye, mucous membrane, or skin - including punctures, such as needlesticks. The Bloodborne Pathogens Standard specifies measures--such as careful handling of contaminated sharps--to reduce these types of injuries and the risk of infection.⁷



Figure 1.3: Sharps Disposal Container. (Copyright [William Rafti via Wikimedia Commons](#); CC BY 2.5)

The three most common bloodborne pathogens are Hepatitis B virus (HBV), Hepatitis C virus (HCV), and Human Immunodeficiency virus (HIV).

Hepatitis B Virus (HBV)⁸ is spread through blood and other bodily fluids. HBC can cause serious harm to the liver, including:

- Liver inflammation or disease (hepatitis) and/or liver cancer, even death

- Acute (short-term) or chronic (long-term) liver infection
- Cirrhosis of the liver and other serious liver diseases

Recovery from acute infection is possible, but chronic HBV infection requires medication. There is no known cure for HBV, but there is a vaccine that can protect against HBV and the severe consequences of hepatitis B infection.

People with HBV do not always have symptoms, so a blood test is important for diagnosis. Most infected persons who become HBV carriers are asymptomatic. However, asymptomatic HBV carriers can spread the virus to others.

People with HBV who do have symptoms may experience:

- Jaundice or yellowing of the skin
- Tiredness
- Abdominal pain
- Loss of appetite
- Occasional nausea
- Vomiting
- Dark urine

Hepatitis C Virus (HCV)⁹ is spread through blood and other bodily fluids and is another member of the hepatitis virus family that:

- Can cause liver inflammation
- Causes liver disease and/or liver cancer
- Can be acute (short term) or chronic (long term)

More than half the people who become infected with hepatitis C will develop a chronic illness.

HCV is curable with timely medical treatment, but there is no HCV vaccine.

Almost half of people who have HCV are unaware they have it, and many people with HCV don't have symptoms. An asymptomatic person with HDV can spread it to others, so blood tests are important for diagnosis.

People with HCV who do have symptoms may experience:

- Jaundice or yellowing of the skin
- Tiredness
- Abdominal pain
- Loss of appetite
- Occasional nausea
- Vomiting
- Dark urine
- Flu-like symptoms

Human Immunodeficiency Virus (HIV)¹⁰ is spread through blood and other bodily fluids. HIV attacks and weakens the body's immune system, making it difficult to fight off common diseases and increasing the chances of developing severe infections or illnesses. Advanced stages of HIV can develop into acquired immunodeficiency syndrome (AIDS).

Many tests are available to diagnose HIV, but there is no vaccine for HIV. There is no cure for HIV or AIDS, although long-term drug therapy can control the disease.

A person can carry HIV for many years without symptoms. HIV might or might not develop into AIDS. Without treatment, people with AIDS typically survive about three years.

Early HIV symptoms resemble flu-like symptoms, including fever, chills, and body aches.

After the virus multiplies in the body, the following symptoms might occur:

- Swollen lymph nodes
- Fatigue
- Diarrhea
- Persistent fever
- Persistent dry cough

Laboratory Safety Guidelines

General Rules:

- No guests are allowed in the lab.
- Know emergency procedures, use and know the location of emergency equipment (fire extinguishers, fire blanket, eye wash station, first aid kit, and broken glass container) and emergency exits.
- In case of fire, evacuate the room and assemble outside the building.
- Report all accidents, no matter how insignificant they appear, to a laboratory supervisor
- Be aware of your surroundings and potential dangers created by others.

Personal Protection:

- Do not smoke, eat, drink, chew gum, or apply cosmetics in a laboratory. Food and drink cannot be stored in lab refrigerators or in the lab.
- When handling hazardous chemicals, blood, body fluids, or other potentially hazardous/contaminated material, wear protective clothing such as long pants, closed-toe and heel shoes, a lab coat, safety glasses, and gloves.
- Shoes should be closed-toe and heel with a slip-resistant tread pattern.
- Gloves will be worn when having contact with hazardous chemicals, blood or other potentially infectious materials, when collecting blood specimens, or when touching contaminated items or surfaces. Discard gloves if torn, punctured, or contaminated.
- Safety glasses or goggles are required when splashes, sprays, or droplets of blood, other infectious materials, or hazardous chemicals may be generated.
- Face shields are required when mixing or pouring hazardous or toxic chemicals.
- Countertop shields can be used in place of face shields when opening blood tubes or when sprays or splashes may be generated.
- Lab coats, which are impervious to blood (may be disposable) and body fluids, are required when working in the laboratory. The coats must be snapped closed for complete protection. Lab coats must be replaced if torn or contaminated.
- Lab coats may not be worn outside of the immediate work area (hallway, restroom, etc.).
- Tie long hair up or back behind the shoulders.
- Do not wear long, dangling jewelry or scarfs.
- Cover cuts or scrapes with a sterile, waterproof bandage before entering a lab.
- Wash hands before and after any laboratory procedure, before leaving laboratory, after removing gloves, and after contamination.
- All work surfaces must be decontaminated after completion of procedures, after any spill of potentially infectious material, and at the end of each lab session.
- Let your lab instructor know before leaving the classroom.

Handling Chemicals and Potentially Infectious Materials:

- Label all reagents and other chemicals clearly with the contents, date of receipt or preparation, and expiration date if applicable.
- Read labels carefully. Be aware of hazardous chemicals and precautions for safe use.
- Keep all containers capped with the appropriate lid.
- Follow all safety procedures for cleaning up spills, splashes, and broken glass. Use broom and dustpan to pick up broken glass.
- Blood specimens and other potentially infectious materials must be disposed of in proper containers.
- Follow instructions about proper disposal of lab reagents.

Handling Equipment:

- Keep your work area clean and free of unnecessary clutter and personal belongings.
- Follow manufacturer's instructions for all equipment operation and maintenance.
- Do not use laboratory equipment without first receiving instruction in its use.
- Notify lab supervisor of malfunctioning equipment.
- Use the biological safety or fume hood as the procedure requires.
- Contaminated sharps are placed in appropriate sharps containers. Recapping of needles is prohibited.

Additional Laboratory Safety Equipment

Fume Hood¹¹

The fume hood is often the primary control device for protecting laboratory workers when working with flammable and/or toxic chemicals.



Figure 1.4: Chemical Fume Hood. (Copyright [Occupational Safety and Health Administration, U.S. Department of Labor](#); Public Domain)

Biological Safety Cabinet

The biological safety cabinet is an important containment device when working with potentially infectious materials. They are designed to protect the laboratory worker against exposure to aerosols or particulates.

Eyewash Station¹²

If your nose, mouth, or eyes have been exposed to blood or body fluids, follow the steps below:

1. Rinse the affected area with large amounts of water or saline solution for several minutes.
2. If running water is not available, use another source of water to rinse, such as bottled water.



Figure 1.5: Eyewash Station. (Copyright [U.S. Centers for Disease Control and Prevention](#); Public Domain)

Infection Control

Healthcare-associated infections are those that are acquired in a health care setting by health care workers or patients. The infection control department within healthcare settings works to lower the incidence of these types of infections.

Chain of Infection Components¹³

They include:

- **Microorganisms:** Disease producing, also called pathogens
 - Virus, parasite, fungus, bacterium
 - Risk factors: Virulence, pathogenicity, ability to enter host
- **Reservoir/Source:** Environment/habitat where a pathogen can live and multiply
 - Environmental surfaces/equipment, body fluids (blood, saliva), urine/fecal material, food/water, soil, skin, respiratory tract
- **Portal of Exit:** How the pathogen exits or leaves reservoir
 - Skin to skin, skin to surface, blood, mucous membranes, oral cavity, fecal
 - Other potentially infectious material (OPIM): Seminal fluid, joint fluid, saliva, urine/fecal material, any body fluid contaminated with blood
- **Modes of Transport:** How a pathogen moves from reservoir to susceptible host
 - *Direct Transmission:* Airborne, droplet, contact (e.g., skin), bite, needlestick or other sharps injury
 - *Indirect Transmission:* Fomites – contaminated equipment or medication, vectors, food, water

- **Portal of Entry:** Opening where the pathogen may enter
 - Body openings (e.g. mouth, eyes, urinary tract, respiratory tract), incisions, wounds
- **Susceptible Host:** The person at-risk: patient or healthcare worker
 - Factors affecting susceptibility (e.g. age, health, co-morbidities, immune system, nutrition, infective dose, medications)

Chain of Infection

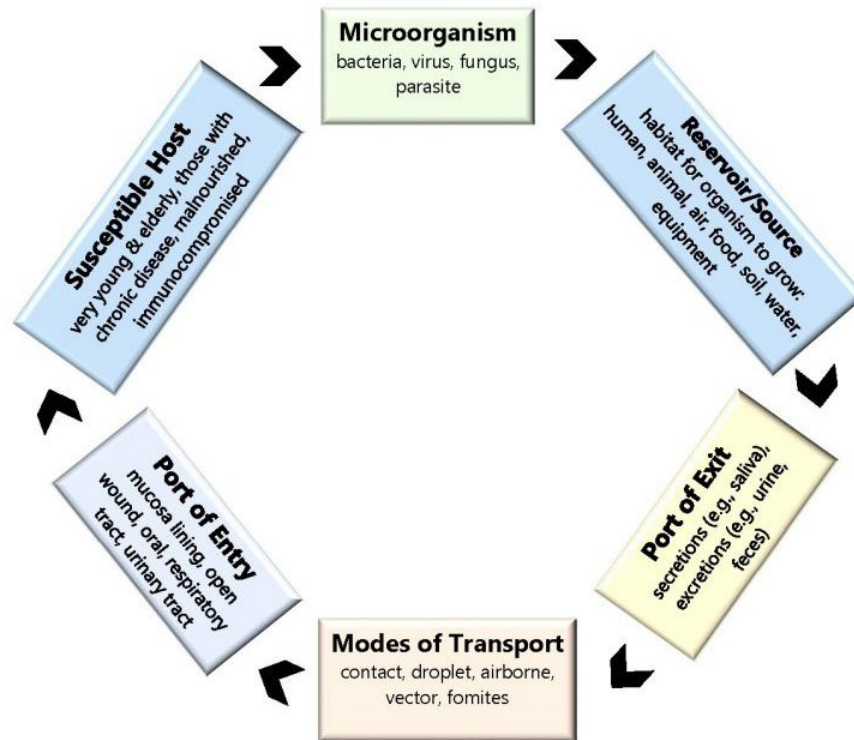


Figure 1.6: Chain of Infection. (Copyright [The National Institute for Occupational Safety and Health](#); Public Domain)

To break the chain of infection, use standard precautions, hand hygiene, and wear PPE. Patients who are at risk of spreading or contracting infections can also be isolated. Isolation is the separation of an infection source from susceptible hosts. Reverse or protective isolation is used for immunocompromised patients.

Infection Prevention for Healthcare Providers¹⁴

Healthcare providers can perform basic prevention measures to prevent infection. There are two tiers of recommended precautions to prevent the spread of infection in healthcare settings: standard precautions and transmission-based precautions.

1. Standard Precautions Used for All Patient Care

Standard Precautions are used for all patient care. They're based on a risk assessment and make use of common sense practices and personal protective equipment use that protect healthcare providers from infection and prevent the spread of infection from patient to patient.

Hand Hygiene protects both healthcare personnel and patients. Hand hygiene means cleaning your hands with:

- Handwashing with water and soap (e.g., plain soap or with an antiseptic).
- Antiseptic hand rub (alcohol-based foam or gel hand sanitizer).
- Surgical hand antisepsis.

Cleaning your hands reduces:

- The potential spread of deadly germs to patients.
- The spread of germs, including those resistant to antibiotics.

- The risk of healthcare personnel colonization or infection caused by germs received from the patient.

Know when to clean your hands:

- Immediately before touching a patient.
- Before performing an aseptic task such as placing an indwelling device or handling invasive medical devices.
- Before moving from work on a soiled body site to a clean body site on the same patient.
- After touching a patient or patient's surroundings.
- After contact with blood, body fluids, or contaminated surfaces.
- Immediately after glove removal.

Know when to use alcohol-based hand sanitizer (ABHS) versus soap and water:

Unless hands are visibly soiled, alcohol-based hand sanitizer (ABHS) is preferred over soap and water in most clinical situations because it:

- Is more effective at killing germs on hands than soap.
- Is easier to use when providing care, especially when moving from soiled to clean activities on the same patient or when moving between care of patients in shared rooms.
- Results in improved skin condition with less irritation and dryness than soap and water.
- Improves hand hygiene adherence.

When to wash with soap and water:

- When hands are visibly soiled.
- Before eating.
- After using the restroom.
- During the care of patients with suspected or confirmed infection during outbreaks of *C. difficile* and norovirus.

Know how to use alcohol-based hand sanitizer (ABHS):

1. Put product on hands and rub hands together. The efficacy (effectiveness) of alcohol-based hand sanitizer depends on the volume applied to the hands. Use the right amount of alcohol-based hand sanitizer product to clean your hands.
2. Cover all surfaces and rub until hands feel dry. This should take around 20 seconds.
3. Pay attention to the areas providers frequently miss:
 - Thumbs
 - Fingertips
 - Between fingers

Know how to wash hands with soap and water:

1. Wet hands with water.
2. Apply the manufacturer recommended amount of product to your hands.
3. Rub hands together vigorously for at least 15 seconds, covering all surfaces of the hands and fingers.
4. Rinse hands with water and use disposable towels to dry. Use a towel to turn off the faucet.
5. Avoid using hot water to prevent drying of the skin.



Figure 1.7: Handwashing. (Copyright [U.S. Centers for Disease Control and Prevention](https://www.cdc.gov); Public Domain)

In addition to hand hygiene, standard precautions include:

- Use personal protective equipment (PPE) whenever there is an expectation of possible exposure to an infectious material.

- Properly handle, clean and disinfect patient care equipment and instruments/devices. Clean and disinfect the environment appropriately.
- Promptly clean and decontaminate spills of blood and other potentially infectious materials. Discard blood-contaminated items in compliance with federal regulations.

2. Transmission-Based Precautions Used for Patients Who May Be Infected or Colonized With Certain Germs

There are also transmission- and germ-specific guidelines providers can follow to prevent transmission and healthcare-associated infections from happening.

Transmission-Based Precautions are the second tier of basic infection control and are to be used in addition to Standard Precautions for patients who may be infected or colonized with certain infectious agents for which additional precautions are needed to prevent infection transmission.

Use **Contact Precautions** for patients with known or suspected infections that represent an increased risk for contact transmission.

- Ensure appropriate patient placement in a single patient space or room if available. In ambulatory settings, place patients requiring contact precautions in an exam room or cubicle as soon as possible.
- Use personal protective equipment (PPE) appropriately, including gloves and gown. Wear a gown and gloves for all interactions that may involve contact with the patient or the patient's environment. Donning PPE upon room entry and properly discarding before exiting the patient room is done to contain pathogens.
- Limit transport and movement of patients outside of the room to medically-necessary purposes.
- Use disposable or dedicated patient-care equipment (e.g., blood pressure cuffs). If common use of equipment for multiple patients is unavoidable, clean and disinfect such equipment before use on another patient.
- Prioritize cleaning and disinfection of the rooms of patients on contact precautions.

For Review: [Contact Precautions Sign from the CDC](#)

Use **Droplet Precautions** for patients known or suspected to be infected with pathogens transmitted by respiratory droplets that are generated by a patient who is coughing, sneezing, or talking.

- Source control: put a mask on the patient.
- Ensure appropriate patient placement in a single room if possible.
- Use personal protective equipment (PPE) appropriately. Don mask upon entry into the patient room or patient space.
- Limit transport and movement of patients outside of the room to medically-necessary purposes.

For Review: [Droplet Precautions Sign from the CDC](#)

Use **Airborne Precautions** for patients known or suspected to be infected with pathogens transmitted by the airborne route (e.g., tuberculosis, measles, chicken pox, disseminated herpes zoster).

- Source control: put a mask on the patient.
- Ensure appropriate patient placement in an airborne infection isolation room (AIIR) constructed according to the Guideline for Isolation Precautions.
- Restrict susceptible healthcare personnel from entering the room of patients known or suspected to have measles, chickenpox, disseminated zoster, or smallpox if other immune healthcare personnel are available.
- Use personal protective equipment (PPE) appropriately, including a fit-tested NIOSH-approved N95 or higher level respirator for healthcare personnel.
- Limit transport and movement of patients outside of the room to medically-necessary purposes.
- Immunize susceptible persons as soon as possible following unprotected contact with vaccine-preventable infections (e.g., measles, varicella or smallpox).

For Review: [Airborne Precautions Sign from the CDC](#)

References

¹ [Laboratories](#) by [Occupational Safety and Health Administration](#), U.S. Department of Labor; Public Domain

² [Hazard Communication](#) by [Occupational Safety and Health Administration](#), U.S. Department of Labor; Public Domain

- ³ [Personal Protective Equipment](#) by [Occupational Safety and Health Administration](#), [U.S. Department of Labor](#); Public Domain
- ⁴ [Portable Fire Extinguishers](#) by [Occupational Safety and Health Administration](#), [U.S. Department of Labor](#); Public Domain
- ⁵ [Fundamentals of Bloodborne Pathogens](#) by [U.S. Centers for Disease Control and Prevention](#); Public Domain
- ⁶ [OSHA Factsheet: OSHA's Bloodborne Pathogens Standard](#) by [Occupational Safety and Health Administration](#), [U.S. Department of Labor](#); Public Domain
- ⁷ [OSHA Factsheet: Protecting Yourself When Handling Contaminated Sharps](#) by [Occupational Safety and Health Administration](#), [U.S. Department of Labor](#); Public Domain
- ⁸ [Fundamentals of Bloodborne Pathogens](#) by [U.S. Centers for Disease Control and Prevention](#); Public Domain
- ⁹ [Fundamentals of Bloodborne Pathogens](#) by [U.S. Centers for Disease Control and Prevention](#); Public Domain
- ¹⁰ [Fundamentals of Bloodborne Pathogens](#) by [U.S. Centers for Disease Control and Prevention](#); Public Domain
- ¹¹ [OSHA Factsheet: Chemical Fume Hoods](#) by [Occupational Safety and Health Administration](#), [U.S. Department of Labor](#); Public Domain
- ¹² [Job Aid: Blood and Body Fluid Exposure](#) by [U.S. Centers for Disease Control and Prevention](#); Public Domain
- ¹³ [Chain of Infection Components](#) by [The National Institute for Occupational Safety and Health \(NIOSH\)](#), [U.S. Centers for Disease Control and Prevention](#); Public Domain
- ¹⁴ [Infection Control Basics](#) by [U.S. Centers for Disease Control and Prevention](#); Public Domain

Except where otherwise noted, content on this page was adapted from:

- [1.1: Lab Safety and Laboratory Notebook](#)(opens in new window) by [Orange County Biotechnology Education Collaborative](#); Public Domain

Images used on this page:

- Flammable substances sign via Wikimedia Commons; public domain
- Explosive substances sign via Wikimedia Commons; public domain
- Gas pressure hazards sign via Wikimedia Commons; public domain
- Corrosive substances sign via Wikimedia Commons; public domain
- Oxidizing substances sign via Wikimedia Commons; public domain
- Toxic substances sign via Wikimedia Commons; public domain
- Health hazard sign via Wikimedia Commons; public domain
- Hazardous substances sign via Wikimedia Commons; public domain
- Environmental hazard sign via Wikimedia Commons; public domain
- Eyewash sign via Wikimedia Commons; public domain
- Emergency shower sign via Wikimedia Commons; public domain
- Fire blanket sign via Wikimedia Commons; public domain
- Fire extinguisher sign via Wikimedia Commons; public domain
- Ionizing Radiation sign via Wikimedia Commons; public domain
- Laser Radiation sign via Wikimedia Commons; public domain
- Explosive substances sign via Wikimedia Commons; public domain
- High Voltage sign via Wikimedia Commons; public domain
- No eating or drinking sign via Wikimedia Commons; public domain
- Nonpotable water sign via Wikimedia Commons; public domain
- Do not enter sign via Wikimedia Commons; public domain
- No smoking sign via Wikimedia Commons; public domain
- Corrosive Substances sign via Wikimedia Commons; public domain
- Poisonous substances sign via Wikimedia Commons; public domain
- Oxidizing material sign via Wikimedia Commons; public domain
- Highly flammable substances sign via Wikimedia Commons; public domain
- Emergency call button sign via Wikimedia Commons; public domain
- First Aid sign via Wikimedia Commons; public domain

- Automated heart defibrillator sign via Wikimedia Commons; public domain
- Ionizing radiation sign via Wikimedia Commons; public domain
- Use protective eyewear sign via Wikimedia Commons; public domain
- Use protective handwear sign via Wikimedia Commons; public domain
- NFPA 704 warning sign sign via Wikimedia Commons; public domain
- Health Hazard sign via Wikimedia Commons; public domain

This page titled [1: Lab Safety](#) is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by [Kari Christenson, Western Technical College](#).

- [1.1: Lab Safety and Laboratory Notebook](#) by [Orange County Biotechnology Education Collaborative](#) is licensed [Public Domain](#).

2: Basic Lab Equipment

Learning Objectives

Course Competency:

Use basic lab equipment.

Learning Objectives:

- Demonstrate pipetting skills
- Use lab glassware
- Operate centrifuges
- Demonstrate use of balances

Introduction

It is important to know the names and functions of different kinds of laboratory equipment so that procedures are done safely and completed successfully. Each piece of laboratory equipment is designed to do a different job. Although some of the equipment may be used in other areas, we will focus on how it is typically used within a clinical lab.

Glassware

Glassware

Laboratory glassware can be made of different quality glass.

- **Flint glass** - inexpensive, disposable. Used in test tubes.
- **Borosilicate glass** - high heat resistance. Used in beakers and flasks. Pyrex and Kimax are brand names.
- **Quartz** - high quality glass used when light transmission is important. Used in cuvettes for spectrophotometers.

Plasticware

Plastics used in the lab can also be made of different types and used at different temperatures. Polyethylene, polystyrene, and polypropylene are examples of plastics used in labware.

Critical vs. NonCritical Measurements

Some labware, such as beakers and graduated cylinders, are used to make estimated measurements. They are termed **noncritical measurements**. When labware is designed to strict standards of accuracy, these are termed **critical measurements**. Volumetric flasks and volumetric pipets are examples of glassware that are used in critical measurements.

Types of Glassware

- **Beakers** – a beaker can hold liquids or solids. They can be used for pouring chemicals into other containers. Heat-resistant beakers can be used for heating liquids.



Figure 2.1: Beaker. (Copyright [Servier Medical Art](#) by Les Laboratoires Servier; [CC BY 4.0](#))

- **Flasks** – there are several types of flasks, including the Erlenmeyer flask. It can be used to hold and heat liquids. This flask has a triangular shape when viewed from the side. The volumetric flask is used for critical measurements and can only measure one volume.

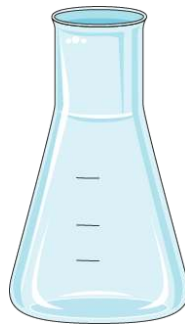


Figure 2.2: Erlenmeyer flask. (Copyright [Servier Medical Art](#) by Les Laboratoires Servier; [CC BY 4.0](#))



Figure 2.3: Volumetric flask. (Copyright [Servier Medical Art](#) by Les Laboratoires Servier; [CC BY 4.0](#))

- **Test tubes** – used to hold smaller amounts of chemicals or patient samples or for biochemical tests. They can be made of glass or plastic and are used frequently in the lab. There are several sizes and shapes of test tubes.

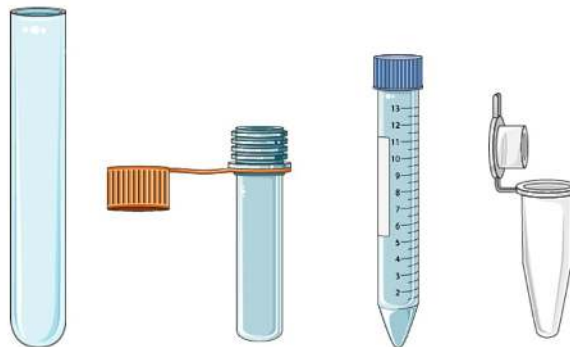


Figure 2.4: Various kinds of test tubes. (Adaptation of Servier Medical Art by Les Laboratoires Servier; [CC BY 4.0](#))

- **Graduated Cylinders** – used to measure liquids. In the clinical setting, they are typically used to measure the volume of 24-hour urine collections or for measuring the total volume of semen. It is important to read the volume level and the bottom of the meniscus (the curved surface of the liquid as it adheres to the sides)

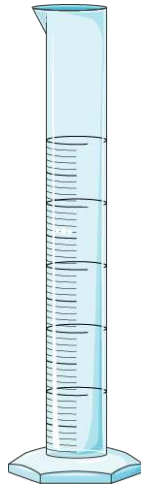


Figure 2.5: Graduated cylinder. (Copyright [Servier Medical Art](#) by Les Laboratoires Servier; [CC BY 4.0](#))

Pipetting

Many sizes and types of pipets are used in the clinical laboratory. It is essential that laboratorians know how to operate different pipettes and be able to select the appropriate pipette for the task being performed.

Pipettes are filled by suction by using a bulb or a mechanical pipette-aid. The upper end of the pipette is fitted to the pipette-aid to draw up the liquid.

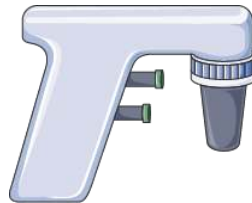


Figure 2.6: Pipette aid. (Copyright [Servier Medical Art](#) by Les Laboratoires Servier; [CC BY 4.0](#))

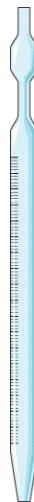


Figure 2.7: Pipette. (Copyright [Servier Medical Art](#) by Les Laboratoires Servier; [CC BY 4.0](#))

TD vs TC

Pipettes are calibrated either **To Deliver (TD)** or **To Contain (TC)**.

To Deliver pipettes are designed to empty by gravity and will deliver the volume marked on the pipette. There will be a small amount of fluid that remains in the tip. If the pipette has a frosted band etched into the top, it is a blowout pipette. This means that the last bit of fluid needs to be expelled from the tip of the pipette. This can be done by using the pipette aid and blowing out the last drop of liquid into the receiving vessel. Most serological pipette have a frosted band and the last drop should be blown out.

Volumetric pipettes typically do NOT have a frosted band and they are calibrated so that the last drop should remain in the tip of the pipette.

To Contain pipettes are calibrated to contain the volume indicated on the pipette. These pipettes must be rinsed to be sure that the appropriate volume is delivered. Micropipettes are examples of TC pipettes.

Serological/Graduated Pipettes

Serological/graduated pipettes are used when more than 1 ml volumes are needed. They can be made of glass or polystyrene plastic. These pipettes are marked in uniform increments. They can transfer the capacity of the pipette or a portion of the total volume. The pipette selected should be closest to the volume needed in the task. For example, if you need to measure 4 ml and there is a 5 ml and a 10 ml serological pipette available, you should select the 5 ml pipette to measure the 4 ml.

Operation:

1. Attach suction aid
2. Aspirate liquid into pipette to the line of the volume required.
3. With a tissue, wipe excess fluid on outside of pipette taking care not to remove any from the tip
4. Holding the pipette near vertical, place in receiving vessel with tip placed against the inner surface
5. Release suction and allow liquid to flow into the container.
6. The last remaining drop is forced out using the pipette aid.



Figure 2.8: Serological pipette. (Original image; Public Domain)

Volumetric Pipettes

Volumetric pipettes are used when critical measurements are required. They are only able to measure one volume of liquid. They have a unique shape with a bulb in the center.

Operation:

1. Attach suction aid
2. Aspirate liquid into pipette to line above the center bulb
3. With a tissue, wipe excess fluid on outside of pipette taking care not to remove any from the tip
4. Holding the pipette near vertical, place in receiving vessel with tip placed against the inner surface
5. Release suction and allow liquid to flow into the container
6. Leave tip in contact with the surface for a few seconds to allow complete drainage
7. A small drop will remain in the tip of the pipette



Figure 2.9: Volumetric pipette. (Original image; Public Domain)

Transfer/Disposable Pipettes

Transfer pipettes are used to dispense small amounts of liquids. They are typically disposable and plastic. They are also used when drops of sample are required.



Figure 2.10: Transfer/disposable pipette. (Copyright [Servier Medical Art](#) by Les Laboratoires Servier; [CC BY 4.0](#))

Introduction to Micropipetting

The ability to measure very small amounts, **microliters (μl)**, of liquid reagents is a fundamental skill needed in the laboratory. Scientists use a device called a **micropipette** to measure these very small volumes with **accuracy**.



Figure 2.11: Micropipette. (Copyright [Servier Medical Art](#) by Les Laboratoires Servier; [CC BY 4.0](#))

Choosing and Setting the Micropipette

There are several sizes of micropipettes used in the clinical laboratory. Micropipettes can broadly be divided into two groups: **variable** and **fixed volume**.

Variable volume micropipettes allow the laboratorian to adjust the volume within a certain range. Common variable micropipettes in the clinical laboratory are labeled 1000 μl , 100 μl , 50 μl , 20 μl , and 10 μl . They can be adjusted up to the volume listed on them. For example, the 1000 μl pipette can measure between 100-1000 μl . The 100 μl pipette can measure between 10 and 100 μl , the 10 μl can measure between 1 and 10 μl , etc. It is important to select a pipette that is the closest to the volume you want to measure. For example, if you were asked to measure 15 μl , you should use the 20 μl variable pipette adjusted to a volume of 15 μl . Even though the 50 μl and the 100 μl can be adjusted to a volume of 15 μl , the pipette that should be selected is the 20 μl .

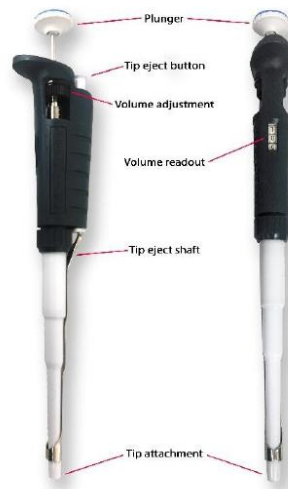


Figure 2.12: Labeled parts of a micropipette, front and back. (Copyright [Orange County Biotechnology Education Collaborative](#) in [1.3: Micropipetting](#); Public Domain)

To adjust the volume of a variable micropipette, you utilize the dial. Some micropipettes require that you push in a small button while turning the dial. There should be no resistance when attempting to turn the dial and adjust the pipet.

Another type of micropipette is the **fixed volume micropipette**. These pipettes can only measure one volume. In a lab, there are typically a set of these types of pipettes.



Figure 2.13: Fixed volume micropipettes in a rack. (Original image; Public Domain)

Both variable and fixed volume micropipettes utilize a disposable tip. It is important to use the correct tip for the pipette you are using.



Figure 2.14: Micropipette tips. (Copyright Nadine90, [CC BY-SA 3.0](#), via [Wikimedia Commons](#))

Basic Operation of a Micropipette

1. Adjust volume of pipette by turning the dial.
2. Hold the micropipette in your dominant hand, and securely place the end into the proper size tip. Once the tip is on, be careful not to touch the tip to anything.
3. With your nondominant hand, hold the container of the liquid that you are needing to measure at eye level.
4. Push the micropipette operating button down to the first stop and hold your thumb in this position.
5. Place the tip of the pipette into the liquid. Be sure that the tip is below the surface of the liquid (approximately 1 cm) so that you don't draw up any air.
6. Gently release your thumb from the plunger to draw fluid into the tip.
7. Examine the tip to confirm that the tip has liquid and that no bubbles are present. If there is liquid on the exterior of the tip, carefully wipe the tip with a Kim wipe. Do not hold the wipe near the tip as this will draw out some of the fluid and make your measurement inaccurate.
8. Place the original container back into the rack.
9. Dispense the liquid in the tip into the receiving vessel by slowly pushing the button to the first stop. After a brief pause (1 second), press the button to the second stop. This forces the last bit of liquid out of the pipette tip (blow-out).
10. Dispose of the tip. Some pipettes do this with a button and with others you must remove the tip manually.

Pipetting Tips¹

- **Pre-Rinse** - Aspirate and then expel the liquid into the pipette tip several times.
- **Consistency** - Maintain consistent rhythm, speed, and technique when depressing and releasing the operational button.
- **Angle** - Keep the angle of immersion as close to vertical as possible. Pull the pipette straight out of the container being sure not to touch the sides.
- **Tip Immersion Time and Aspiration Rate** - Be sure to keep the pipette tip immersed in the liquid for enough time to allow the tip to fill. This is especially important for viscous liquids.
- **Depth** - The correct tip immersion depth improves pipetting accuracy. If the tip is immersed too far, more liquid is aspirated. If the tip is not immersed far enough, air can be drawn in, resulting in inaccurate volume. Resting the tip against the walls or bottom of the container restricts entry of the sample.

Table 2.1: Relationship between Pipette Volume and Immersion Depth²

Pipette Volume	Immersion Depth
2 and 10 μ l	1 mm
20 and 100 μ l	2-3 mm
200 and 1000 μ l	3-6 mm
5000 μ l and 10 ML	6-10 mm

Watch "[How to pipette correctly – a short step-by-step introduction into proper pipetting](#)" by Eppendorf for demonstration of effective pipetting techniques.

Additional tips available at "[The Fundamentals of Pipetting](#)" by ThermoFisher Scientific [[archived link here](#)].

Centrifuge

Centrifuges are common pieces of equipment used in laboratories. They use centrifugal force to separate substances based on density. Fluids such as blood or urine are spun at a high speed within a container. Tubes spin around a central rotor. It is critical that the load is balanced before starting the centrifuge. Tubes of the same size and equal volume must be placed across the rotor from one another. Tubes should remain capped and the lid of the centrifuge must remain closed for the entire time. The speed (revolutions per minute or RPM) and the time of centrifugation will change depending on the samples being spun/separated. For example, urine and blood samples require different speeds and times. Follow the procedure at your institution. To properly care for this important piece of equipment, maintenance needs to be performed. This includes checking the rotation speed, checking the timer, and promptly cleaning up any spills.

There are several types of centrifuges used in the clinical lab:

- **Clinical Centrifuge** – General centrifuge used for spinning blood and urine samples

- **Serological Centrifuge** – Smaller centrifuge used in blood banking to spin smaller test tubes
- **Microcentrifuge** – Used to spin small microcentrifuge tubes at high speed

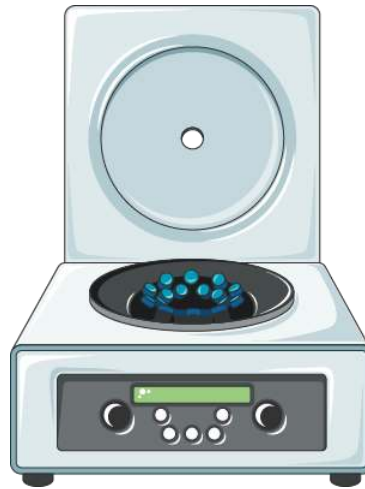


Figure 2.15: Clinical centrifuge. (Copyright [Servier Medical Art](#) by Les Laboratoires Servier; [CC BY 4.0](#))

Balance

Balances are another piece of equipment used in laboratories although not nearly as commonly as a centrifuge. They are used to measure the weight of a substance. Always follow the manufacturer's instructions for operation. In general, keep balances clean and protect them from jarring. Keep in a location free from drafts and vibration. Ensure that the balance reads zero when the weighing area is empty. Tare (set the balance to zero) with the empty weighing container on the weighing pan.

References

¹ Pipetting tips include original content and content adapted from Pushparaj P. N. (2020). Revisiting the Micropipetting Techniques in Biomedical Sciences: A Fundamental Prerequisite in Good Laboratory Practice. *Bioinformation*, 16(1), 8–12. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6986936/>. [CC BY 4.0](#).

² Mettler-Toledo (n.d.). "Rainin Classic Manual Pipette: Operating Instructions." *MT.com*. https://www.mt.com/dam/RAININ/PDFs/UserManuals/rainin_classic_manual_EN.pdf

Except where otherwise noted, content on this page includes original content and content adapted from:

- **1.3: Micropipetting** by [Orange County Biotechnology Education Collaborative](#); [CC BY 4.0](#).

This page titled [2: Basic Lab Equipment](#) is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by [Kari Christenson](#), Western Technical College.

- **1.3: Micropipetting** by [Orange County Biotechnology Education Collaborative](#) is licensed [CC BY 4.0](#).

3: Microscope

Learning Objective

Course Competency:

Use basic lab equipment.

Introduction

An important piece of equipment used in several departments in a clinical laboratory is the microscope. In the hematology department, microscopes are used to examine stained blood smears. Stained microorganisms are examined in the microbiology department. In urinalysis, the microscope is used to view urine sediment. The microscope is a delicate and expensive piece of equipment, so care must be taken to use it properly.

Clinical microscopes are light microscopes. The microscope uses bright light to illuminate through the specimen and provides an inverted image at high magnification and resolution. There are two lenses that magnify the image of the specimen – the objective lens on the nosepiece and the **ocular lens** (or eyepiece). To determine the **total magnification** of the specimen, you must multiply the objective lens magnification with the ocular lens magnification.

When viewing unstained cells, the use of phase-contrast is helpful. A compound brightfield microscope can be fitted with a special objective and phase condenser. In phase contrast microscopy, the background looks grey, and the specimen is bright. In the lab, urine sediment is visualized using phase contrast.

A third type of microscope, called an epi-fluorescent microscope, can also be used in clinical labs. These microscopes use ultraviolet light to view the specimen. The specimens need to be stained with a fluorescent dye in order to visualize them.

Microscope Parts

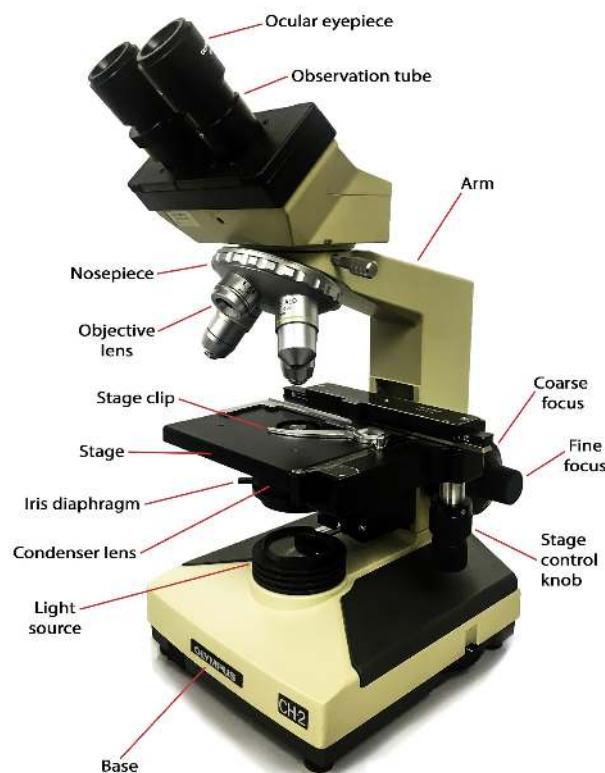


Figure 1. Labeled parts of a microscope.

Oculars

Clinical microscopes are binocular, which means they have two eyepieces. They are designed for the user to utilize both eyes when viewing the specimen. This helps to prevent eyestrain. The oculars are adjustable to fit different eye distances (interpupillary distance). The eyepieces, or oculars, are attached to a tube that is connected to the arm of the microscope. There is a lens in each ocular. Typically, the magnification of the ocular lens is 10x.

Objective Lenses

On the underside of the microscopes arm, is a revolving nosepiece. This part of the microscope holds the objective lenses. The three most common objective lenses used in clinical labs are the low power, high power, and oil immersion objectives. The low power objective is typically 10x magnification. The high-power objective is 40x and the oil immersion objective is 100x. Each objective is marked with a colored band and the magnification. As the magnification of the objective increases, so does the overall length of the tube. Recall, to determine the total magnification, you must multiply the magnification of the ocular lens by the magnification of the objective lens.

Table 3.1: Magnification

Objective	Objective Magnification	Colored Band	Ocular Magnification	Total Magnification
Low power	10x	Yellow	10x	100x
High power	40x	Blue	10x	400x
Oil immersion	100x	Black	10x	1000x

The oil immersion objective is the only objective that requires the use of oil. Oil is placed between the slide and the objective. This prevents air from refracting the light out of focus on this very magnified lens. The high power (40x) objective is long enough to touch any oil that is left on a slide, but this objective is not designed to be in contact with oil. Any oil on the 40x objective should be cleaned off with lens cleaner and lens paper. The oil immersion objective is the only one that should ever have oil on it.

Stage

The stage is located under the revolving nosepiece with the objective lenses. The slide with the specimen is placed on the stage. It is held in place with the stage clip. The slide is moved with the stage control knobs.

Iris Diaphragm

Lever that opens and closes to control the amount of light reaching the specimen.

Condenser

The condenser focuses the available light through the specimen and into the objective lens. It enhances specimen contrast. The condenser can be raised up to the specimen when more light is needed and can be lowered away from the specimen when less light is needed.

Light Source

The slide is illuminated by the light source in the base of the microscope. Most newer microscopes have LED lights that do not need to be replaced.

Light Intensity Knob

The light intensity knob, also called a rheostat, controls how bright the light bulb is. It's like a dimmer switch in your house!

Coarse and Fine Focus Adjustment Knobs

The course and fine focus adjustment knobs are typically on the side of the microscope base. The course adjustment is used only with the low power objective. The fine focus is used to sharpen the focus of the image after the object is found with the course adjustment on low power. The fine focus knob is used for the high power and oil immersion objectives. Use of the course adjustment knob with these objectives causes too much movement and may break the slide. Microscopes are typically parfocal so once the field is in focus on low power, only fine focus adjustments are necessary when switching to higher power objectives.

It is very important to learn how to handle and use a microscope properly. Review the following rules and tips for using and handling your microscope.

General Procedure

1. Always use one hand around the microscope arm and one hand under the microscope base.
2. Carry it in a vertical position without swinging, tipping, dropping or bumping the microscope.
3. Place the microscope gently on the lab bench. Never slide it across the table.
4. Clean all lenses with lens cleaner and lens paper. Never use paper towels or Kimwipes.
5. Carefully plug in the microscope.
6. Turn on the microscope and rotate the **nosepiece ring (turret)** to click the **10x objective** lens in place. Do not use the objective lens to rotate!
7. Turn the **light control (rheostat)** halfway to adjust the amount of light.
8. On the side of the microscope are two knobs, one on top of the other. The larger of the two knobs is the **coarse focus adjustment knob**. Turn the knob so that the stage goes down as far as it can.
9. Obtain a slide.
10. Place the slide on the stage (specimen side up) and secure it with the **stage clip**.
11. Use the coarse focus knob to move the stage as high as it can go.
12. Use **stage adjustment knobs** to center the specimen so that the light from the **light source** can pass through it.
13. Looking through the ocular lenses, adjust them to fit your eye span (interpupillary distance). Only one image should be seen and both eyes need to be open. Then lower the stage with the coarse focus adjustment knob until the specimen comes into view.
14. Use the **fine focus adjustment knob** to make the image as clear as you can.
15. At this point there are different adjustments you can make to improve the quality of the image:
 1. The **rheostat** on the side of the microscope controls the intensity of the light. If it is too bright or dim at any time, use this knob to adjust the light.
 2. The **condenser** will also adjust the light intensity. The condenser gathers and focuses the light to illuminate the specimen. Move the condenser with the condenser adjustment knob so that it approaches the stage. It should almost be touching the bottom of the slide.
 3. The **iris diaphragm** adjusts the aperture of the opening and controls the amount of light that exits the condenser (or illuminates the specimen). You can open and close this aperture, for most purposes it should be fully open, but sometimes partially closing it will increase contrast in the image.
16. Once the specimen is in focus, turn the nosepiece to the 40x objective lens and click it into place. Use the fine focus to make the image clear. (**NEVER** use the coarse focus at this or any higher magnification or you risk snapping the slide or worse, snapping the lens!!!) If needed, make any light adjustments with the rheostat, condenser, or diaphragm.
17. To use the oil immersion lens, rotate the nosepiece **BETWEEN** the 40x and the 100x lenses so that the wand or dropper containing the oil can reach the slide. Place a generous drop of oil on the slide and click the 100x objective lens into place. The lens will slide into the drop of oil.
18. Use the fine focus to make the image clear. If needed, make light adjustments with the rheostat, condenser, or diaphragm.
19. **NEVER return to the 40X objective lens after there is oil on the slide.** If you are having trouble focusing using the oil immersion lens, you must go back and use the 10x lens to re-center (turn the nosepiece so that the 40x objective lens is NOT dragged through the oil on the slide). Then go directly back to the 100x lens. If this doesn't work, the slide must be wiped clean and you should start over.
20. When finished with the slide, lower the stage and remove the slide. (Do not lower the stage if you are going to view a different slide). Clean the oil off the slide.
21. Clean the oil immersion objective with lens paper and lens cleaner to remove all oil.
22. Clean the oculars and objectives with the clean lens paper and lens cleaner.
23. Turn off the microscope light and unplug the microscope.
24. Position the nosepiece in the lowest position with the coarse adjustment.
25. Center the stage so it doesn't stick out on the sides.
26. Cover the microscope and return to storage.

General Rules/Procedural Reminders

- Carry the microscope with two hands - one under the base and the other holding the arm.
- Never turn the nose piece by the objective lens.
- Use only lens paper to clean microscope lenses (oculars and objectives). Other materials will scratch.
- Only use immersion oil with the oil-immersion objective.
- Avoid jarring or bumping the microscope.
- Use the coarse adjustment only with the low power objective.

References

Content on this page was adapted from:

- [1.5: Microscopy](#) by [Orange County Biotechnology Education Collaborative](#); [Public Domain](#)

This page titled [3: Microscope](#) is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by [Kari Christenson, Western Technical College](#).

- [1.5: Microscopy](#) by [Orange County Biotechnology Education Collaborative](#) is licensed [CC BY 4.0](#).

4: Medical Terminology

Learning Objectives

Course Competency:

Apply medical terminology to lab procedures and practices.

Learning Objectives:

- Explain why using the correct medical terminology is important
- Identify sources for defining medical terminology
- Define significant medical terms used in a health care setting.

Use of medical terminology is embedded throughout the MLT program. This unit of the Basic Lab Skills course provides a brief introduction to the study of medical terminology, including the components of medical terms, including prefixes, suffixes, and word roots. Additional courses in the program will provide opportunities to expand, deepen, and contextualize this knowledge.

Word Components

Medical terms can be defined by breaking down the term into word components and defining each component. These word components include prefixes, word roots, and suffixes.

- The **prefix (P)** appears at the beginning of the medical term and adds meaning to the word root.
- The **word root (WR)** is the core of many medical terms and refers to the body part or body system to which the term is referring.
- The **suffix (S)** is at the end of the medical term and usually indicates a procedure, condition, or disease. When defining a medical term, start the definition with the suffix. For example, consider the common medical condition *tonsillitis*. The word root “tonsil” refers to the tonsils, an anatomical part of the body found in the throat. The suffix “-itis” refers to inflammation. Therefore, the definition of the medical term *tonsillitis* is “inflammation of the tonsils.”
- **Combining vowel (CV)** is typically the letter “o” that helps with pronunciation. A combining vowel can connect a word root and a suffix or two word roots. When a word root is combined with a combining vowel, the word part is referred to as a **combining form (CF)**. See examples of word components in the table below.

Table 4.1: Components of Medical Terms

Component	Definition	Example
Word Root (WR)	Core of the word	hepat/itis hepat = liver
Prefix (P)	Attached to the beginning of the word root to alter its meaning	sub/hepatic sub- = below
Suffix (S)	Attached to the end of a word root to alter its meaning	hepat/ itis -itis = inflammation
Combining Vowel (CV)	Typically, an “o” is used to assist with pronunciation	hepat/ o /megaly
Combining Form (CF)	Word root with a combining vowel	hepat/o

By defining the meaning of each component of a medical term, it is possible to construct the full meaning of the term based on the meanings of its components. See an example of defining a medical term by its components in the following box.

Defining a Medical Term by its Components

The medical term *arthropathy* contains the following word components:

- The word root “**arthr-**” that refers to a joint
- The combining vowel “**o**” that creates the combining form arthro-

- The suffix “-**pathy**” that refers to a disease

When putting the meaning of these components together, the medical term *arthropathy* means “**disease of a joint.**”

Definitions of common prefixes, word roots, suffixes, and combining vowels will be discussed in greater detail in the following sections of this chapter.

Types of Medical Terms

Medical terms can be divided into three categories of terms, based on the origin of their components:

1. Terms built from word components of Greek and Latin origin
2. Terms NOT built from word components of Greek and Latin origin
3. Terms that look like they are made from word parts but are not easily translated

Definitions of medical terms built from word components of Greek and Latin origin can be easily identified by analyzing the components, defining each component, and then building the meaning of the term based on the meaning of its components.

Medical terms NOT built from word components of Greek and Latin origin, or terms that are not easily translated, must be learned by memorizing and recalling the definitions.

Table 4.2: Types of Medical Terms

Type	Origin	Examples	Methods
Terms That Can Be Easily Built From Word Components	Word components of Greek and Latin origin that are combined to create literal translations	Hepatitis Arthropathy	Definitions are identified by analyzing the word components, defining the word components, and then building the meaning of a medical term based on the meaning of its components
Terms NOT Easily Built From Word Components	Eponyms (i.e., a disease or procedure named after someone who discovered it)	Parkinson’s disease – a disease first described by a physician named James Parkinson	Definitions are learned by memorizing and recalling definitions
	Acronyms (i.e., an abbreviation formed from the initial letters of other words and pronounced as a word)	AIDS – an acronym that stands for the disease called acquired immunodeficiency syndrome	
	Modern language (i.e., terms created in a language currently used)	Post-traumatic stress disorder – a mental health diagnosis created by using the English language	

Defining Medical Terms by Breaking Down Components

Defining medical terms built from word components is easily accomplished by analyzing the components of the term, defining each component, and then building the overall definition of the medical term.

For example, see the table for instructions on how the definition of the medical term “osteopathopathy” can be identified by breaking down its components and defining them.

Table 4.3: Technique for Defining Medical Terms Based on Their Components

Technique	Instructions	Steps
Analyze Components	1. Divide into word parts. 2. Label the word parts. 3. Underline and label each combining form.	1. oste/o/arthr/o/pathy 2. oste = WR/; o = CV/; arthr = WR/; o = CV/; pathy = S 3. <u>oste/o/arthr/o/pathy</u>

Define Components	1. Define each word part in the term.	1. oste = bone; arthr = joint; pathy = disease
Build Definition	1. Place word parts together to define the term. Begin by defining the suffix, and then move to the beginning of the term in the order they appear. 2. Add combining vowels to pronounce the term.	1. Disease of the bone and joint 2. oste/o/arthr/o/pathy

When breaking down medical terms into its components, place slashes between the components and a slash on each side of a combining vowel. Notice how the term is defined by beginning with the meaning of the suffix and then shifts to the beginning of the term with the meaning of the word parts in the order they appear. Additional examples of identifying definitions of medical terms by analyzing their components are provided in the following box.

Additional Examples of Identifying Definitions of Medical Terms by Analyzing Their Components

Example: Osteoarthritis

1. Analyze Components: **oste/o/arthr/itis**
2. Define Components:
oste is a word root (WR) that means “bone”
/o/ is a combining vowel (CV)
arthr is a word root (WR) that means “joint”
-itis is a suffix (S) that means “inflammation”
3. Build Definition: **Inflammation of bone and joint**

Example: Intravenous

1. Analyze Components: **intra/ven/ous**
2. Define Components:
intra- is a prefix (P) that means “within”
ven is a word root (WR) that means “vein”
-ous is a suffix (S) that means “pertaining to”
3. Build Definition: **Pertaining to within a vein**

Language Rules

Language rules are a good place to start when building foundational knowledge of medical terminology. Many medical terms are built from word parts and can be translated literally. At first, literal translations sound awkward. However, after you build a medical vocabulary and become proficient at using it, the awkwardness will slip away. Suffixes will no longer be stated and will be assumed. For example, the definition of *intravenous* can be condensed from “pertaining to within a vein” to “within the vein.” It should be noted that as with all language rules there are always exceptions, often referred to as “rebels.”

Pronunciation

Pronouncing complex medical terms can be challenging. When learning how to pronounce medical terms, special marking above vowels indicates the proper pronunciation. When you see a macron (i.e., straight line) above the vowel, that means the vowel sound is long. A u-shaped symbol above a vowel indicates a short vowel sound. Additionally, capital letters indicate where to place the emphasis when pronouncing a word. See examples of pronunciation markings in Table 4. Try pronouncing each example out loud.

Table 4.4: Pronunciation Guidelines

Guidelines	Example
Markings indicate the proper phonetic sounds.	doctor (dōk-tōr)

A macron (a line above the letter) indicates a long vowel sound. Examples: ā in “play” ē in “bee” ī in “wine” ō in “go” ū in “mule”	prorate (prō-rāt)
A u-shaped mark above the vowel indicates a short sound. Examples: ă in “mad” ĕ in “bet” ĭ in “tip” ŏ in “mop” ŭ in “cup”	medical (mĕd-ĭ-cāl)
Primary emphasis of pronunciation is indicated by capital letters.	debride (di-BRĒD) dehydration (dĕ-hĭ-DRĀ-shŏn)

Categories of Medical Terms

Medical terms can be classified into the following general categories of terms:

- **Anatomical:** Terms used to describe specific areas and movement of the body, as well as the relation of body parts to each other.
 - An example of an anatomical term is *medial*, which describes the middle or direction toward the middle of the body.
- **Disease and Disorder:** Terms that describe deviation from the normal structural or functional state of an organism, generally associated with certain signs and symptoms.
 - An example of a disease is *bronchopneumonia*, an infection of the bronchi in the lungs.
- **Diagnostic/Procedural:** Terms related to the process of identifying a disease, condition, or injury.
 - An example of a diagnostic term is *transrectal ultrasound*, an ultrasound procedure used to diagnose prostate cancer.
- **Therapeutic:** Terms regarding the treating or curing of diseases.
 - An example of a therapeutic term is *nebulizer*, which is a device that creates a mist used to deliver medication for respiratory treatment.

Prefixes

The prefix appears at the beginning of a medical term and adds meaning to the root word, like adjectives add meaning to nouns in the English language.

Prefixes often indicate:

- **Number**, such as *bi-*, meaning “two”
- **Position**, such as *sub-*, meaning “under”
- **Direction**, such as *intra-*, meaning “within”
- **Time**, such as *brady-*, meaning “slow”
- **Negation**, such as *a-* and *an-*, meaning “without”

Here is a list of commonly used prefixes. It is helpful to memorize these common prefixes as you build your knowledge of medical terminology:

- **a-, an-:** Absence of, without, not
- **ab-:** Away from, take away
- **ad-:** Towards, to, near
- **ambi-, ambo-:** Both
- **amphi-:** Double, both sides

- **ana-**: Up, apart
- **ante-**: Before, in front of, forward
- **anti-**: Opposing, against
- **ap-, apo-**: Separated, derived from
- **aut-, auto-**: Self, by yourself
- **bi-**: Double, two, twice, both
- **brachy-**: Short
- **brady-**: Slow
- **cata-**: Lower, down, against
- **circum-**: Around
- **co-, con-, com-**: Together, with
- **contra-**: Against, opposed to
- **de-**: Down, from
- **di-**: Twice, two
- **dia-**: Through, apart, across, between
- **dis-**: Apart from, free from
- **dys-**: Bad, painful, difficult
- **e-, ec-, ex-**: From, away from, out of
- **ect-, exo-, ecto-**: Outside, outer, on
- **em-, en-**: In
- **end-, endo-, ent-, ento-**: Within, inner
- **epi-, ep-**: Upon, over, on
- **eu-**: Normal
- **extra-, extro-**: Beyond, outside of, outward
- **hemi-, semi-**: Half, half of
- **hyper-**: Above, excessive, beyond
- **hyp-, hypo-**: Below, beneath, deficient
- **im-, in-**: Intro, in, within
- **infra-**: Below, beneath
- **inter-**: Between
- **intra-**: Within, inside
- **intro-**: Into, within
- **iso-**: Same, equal
- **macro-**: Large
- **mal-**: Bad
- **mes-, meso-**: Middle
- **meta-**: Beyond, changing
- **micro-, micr-**: Tiny, small
- **mono-, uni-**: One
- **mult-, multi-**: Many, multiple
- **neo-**: New, recent
- **nulli-**: None
- **oligo-**: Small amount
- **ortho-**: Straight, normal
- **pan-**: All
- **para-**: Beyond, beside, after
- **per-**: Through
- **peri-**: Around
- **poly-**: Many, excessive
- **post-**: After, following, behind
- **pre-, pro-**: In front of, before, preceding
- **presby-**: Old age

- **pseudo-**: False
- **quadri-**: Four
- **re-**: Backward, again
- **retro-**: Behind, backward
- **semi-**: Half, partial
- **sub-**: Under, beneath
- **super-, supra-**: Above, excessive, superior
- **sym-, syn-**: With, together
- **tachy-**: Rapid, fast
- **trans-**: Through, across
- **tri-**: Three
- **ultra-**: Excessive, beyond
- **uni-**: One

Examples of Common Prefixes in Medical Terms

- **Bilateral**: Both sides
- **Dysuria**: Painful urination
- **Hypertension**: Excessive/high blood pressure
- **Oliguria**: Small amount of urine production
- **Pericardial**: Around the heart
- **Tachycardia**: Rapid heart rate

Word Roots and Combining Vowels

A word root is the primary building block of a medical term and refers to the body part or body system to which the term is referring. Some words contain more than one word root. The order is generally dictated by common practice. As you practice throughout this course, you will learn more about how to determine the order of word roots. Word roots will be further discussed in more detail in each chapter as they relate to specific body systems.

A combining vowel is a word part, most often the letter “o,” that helps pronunciation.

Here is a list of commonly used word roots and their combining vowel. It is helpful to memorize these components as you build your knowledge of medical terminology.

- **abdomin/o**: Abdomen
- **andr/o**: Male
- **angi/o**: Vessel
- **arteri/o**: Artery
- **arthr/o**: Joint
- **audi/o**: Hearing
- **bronch/i, bronch/o**: Bronchus/lung
- **carcin/o**: Cancer
- **cardi/o**: Heart
- **cholecyst/o**: Gallbladder
- **chondr/i, chondr/o**: Cartilage
- **col/o**: Colon
- **cysti, cyst/o**: Bladder or cyst
- **cyt/o**: Cell
- **duoden/o**: Duodenum
- **encephal/o**: Brain
- **erythr/o**: Red
- **esophag/**: Esophagus
- **gastr/o**: Stomach
- **glyc/o**: Sugar
- **gynec/o**: Female reproductive system

- **hemat/o, hem/o:** Blood
- **hist/o:** Tissue
- **hyster/o:** Uterus
- **laryng/o:** Larynx
- **leuk/o:** White
- **men/o:** Menstruation
- **my/o:** Muscle
- **nephro/o:** Kidney
- **neur/o:** Nerve
- **opt/o:** Vision
- **orch/o:** Testis, testicle
- **pneum/o:** Lungs
- **thorac/o:** Chest

Examples of Common Word Roots in Medical Terms

Cardiomegaly: Enlargement of the heart

Encephalitis: Inflammation of the brain

Gynecologist: Specialist who treats disorders and diseases of female reproductive organs

Pneumonia: Disease state of the lung

Suffixes

The suffix is at the end of the word and adds meaning. When building a definition of a medical term from its components, start with the meaning of the suffix. For example, *hepatitis* is defined as “inflammation of the liver.”

Suffixes often indicate:

- **Procedures:** For example, *-scopy* means “visual examination”
- **Conditions:** For example, *-itis* means “inflammation”
- **Diseases:** For example, *-oma* means “tumor”

Here is a list of commonly used suffixes in medical terms. It is helpful to memorize these common suffixes as you build your knowledge of medical terminology.

- **-ac:** Pertaining to
- **-ad:** Toward
- **-al:** Pertaining to
- **-algia, -dynia:** Pain
- **-ar:** Pertaining to
- **-ary:** Pertaining to
- **-cele:** Protrusion
- **-centesis:** Surgical withdrawal of fluid
- **-cide, -cidal:** Kill, destroy
- **-coccus:** Berry shaped
- **-cyte:** Cell
- **-eal:** Pertaining to
- **-ectomy:** Surgical removal
- **-emia:** Pertaining to blood
- **-esis:** Condition
- **-gen:** Substance that produces/causes
- **-genic:** Produced by
- **-gram:** Record of
- **-graph:** Instrument used to record
- **-graphy:** A process of recording

- **-ia**: Condition of, diseased state, abnormal state
- **-ic, -ior**: Pertaining to
- **-ism**: State of, condition, theory
- **-itis**: Inflammation
- **-logist**: Specialist who studies and treats disorders
- **-logy**: Study of
- **-lysis**: Destruction, breakdown
- **-megaly**: Enlargement
- **-meter**: Instrument used to measure
- **-metry**: Process of measuring
- **-oid**: Resembling
- **-oma**: Tumor
- **-opia**: Vision condition
- **-opsy**: To view
- **-osis**: Abnormal condition
- **-ous**: Pertaining to
- **-pathy**: Disease, abnormality
- **-penia**: Lack of, deficiency
- **-pexy**: Surgical fixation, suspension
- **-phagia**: Eating
- **-phobia**: Intense fear of
- **-plasia**: Formation, development
- **-plasm**: Growth, substance, formation
- **-plasty**: Surgical reconstruction
- **-plegia**: Paralysis
- **-pnea**: Breathing
- **-poiesis, -poietic**: Production, manufacture of
- **-ptosis**: Drooping
- **-rrhage**: Excessive flow
- **-rrhagia**: Excessive flow
- **-rrhaphy**: Suturing in place, fixation
- **-rrhea**: Flow, discharge
- **-rrhexis**: Rupture or breaking away
- **-sclerosis**: Hardening
- **-scope**: Instrument used to visually examine
- **-scopic**: Visual examination
- **-scopy**: Process of visual examination
- **-spasm**: Sudden, involuntary
- **-stasis**: To stop
- **-stomy**: Artificial surgical opening
- **-tome**: Instrument used to cut
- **-tomy**: Process of incision, cutting into
- **-tripsy**: Crushing
- **-trophic, -trophy**: Growth, development

Note there are several suffixes that mean “pertaining to,” such as -ac, -al, -ar, -ary, -eal, -ic, -ior, and -ous.

Examples of Common Suffixes in Medical Terms

- **Arachnophobia**: Fear of spiders
- **Amniocentesis**: Procedure to withdraw fluid from the amniotic sac
- **Anemia**: Low levels of hemoglobin in the blood
- **Cardiomegaly**: Enlargement of the heart
- **Colonoscopy**: Procedure that visually examines the colon

- **Dyspnea:** Difficulty breathing
- **Hemiplegia:** Paralysis of one side of the body
- **Mammography:** Radiological process of examining the breast tissue
- **Myalgia:** Muscle pain

Glossary

Combining form (CF): When a word root (WR) is combined with a combining vowel, the word part is referred to as a combining form.

Combining vowel (CV): A word component, typically the letter “o,” that helps with pronunciation. A combining vowel can connect a word root and a suffix or two word roots.

Prefix (P): A word component that appears at the beginning of the medical term and adds meaning to the word root. Prefixes often indicate number, position, direction, time, or negation.

Suffix (S): A word component at the end of the medical term that usually indicates a procedure, condition, or disease.

Word root (WR): A word component that is the core of many medical terms and refers to the body part or body system to which the term is referring.

Additional Resources

OpenRN. "Blood Terminology," *Medical Terminology 2e*; CC BY 4.0.

OpenRN. "Urinary System Terminology," *Medical Terminology 2e*; CC BY 4.0.

OpenRN. *Medical Terminology 2e*; CC BY 4.0.

References

Except where otherwise noted, all content in this section adapted from:

OpenRN. "Foundational Concepts - Identifying Word Parts," *Medical Terminology 2e*; CC BY 4.0.

This page titled [4: Medical Terminology](#) is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by [Kari Christenson](#), Western Technical College.

- [1.2: Components and Categories of Medical Terms](#) by Open Resources for Nursing is licensed [CC BY 4.0](#).
- [1.3: Common Prefixes](#) by Open Resources for Nursing is licensed [CC BY 4.0](#).
- [1.4: Common Word Roots and Combining Vowels](#) by Open Resources for Nursing is licensed [CC BY 4.0](#).
- [1.5: Common Suffixes](#) by Open Resources for Nursing is licensed [CC BY 4.0](#).
- [1.7: Glossary](#) by Open Resources for Nursing is licensed [CC BY 4.0](#).

CHAPTER OVERVIEW

5: Health Careers

Learning Objectives

Course Competency:

Investigate health career options.

Learning Objectives:

- List departments of the medical laboratory.
- List common laboratory occupations and personnel.
- List responsibilities of common laboratory occupations.
- Describe the educational requirements for common laboratory occupations and personnel.
- Describe career-laddering options for laboratory occupations.
- Examine the relationship of laboratory occupations with other healthcare providers.
- Examine professional organizations and certification agencies

Introduction

Types of Labs

By far the most common type of lab that MLT students find employment in after graduation are in hospital or clinic laboratories.

- **Hospital Labs** – These are laboratories located inside hospitals. They are typically staffed 24 hours/day and offer a wide variety of tests.
- **Clinic Labs** – These are laboratories located inside clinics. They have more restricted staffing hours compared to hospitals. Fewer tests are typically performed in a clinic lab. Some tests are sent out to their associated hospital or to a reference laboratory.

Other types of labs include reference, research, nursing homes, government agencies, industry, and military labs.

Lab Departments

The clinical laboratory (whether physically located in a hospital or clinic) can broadly be divided into two main areas: **clinical pathology** and **anatomic and surgical pathology**.

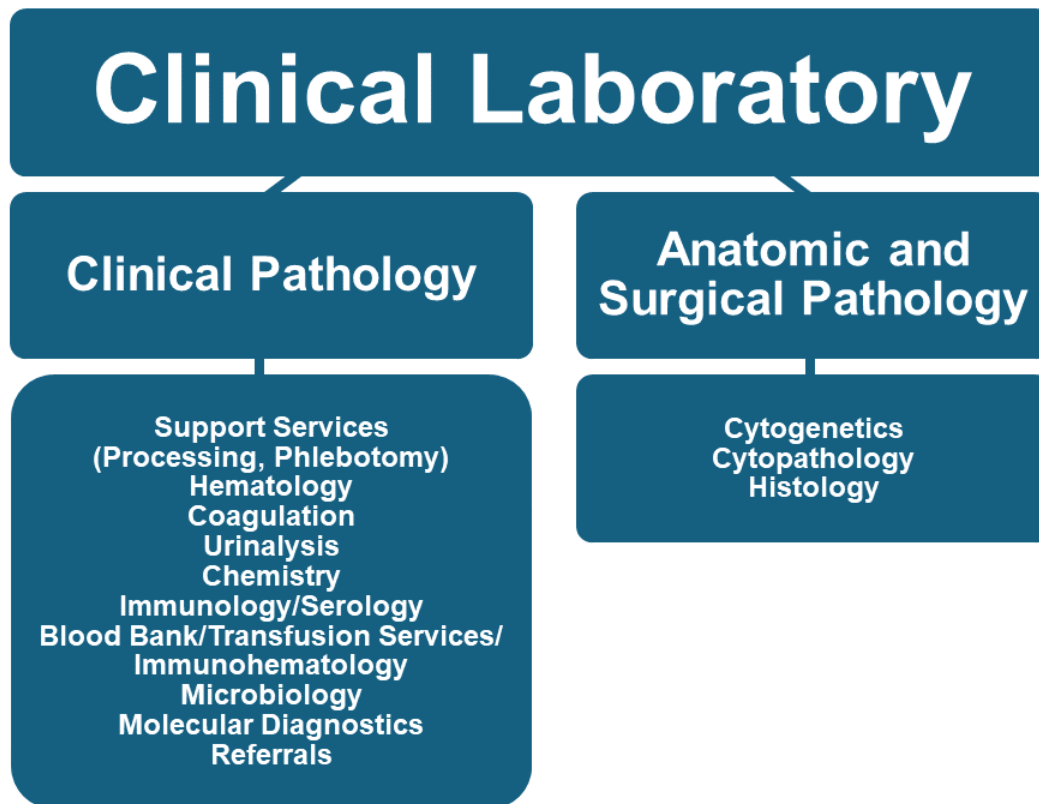


Figure 5.1: Organization of the Clinical Laboratory

Clinical Pathology Departments

In the clinical pathology area, analysis of blood and other body fluids is performed. Depending on the size of the lab, there can be several different areas/departments where testing is performed. This is the area of the lab where Medical Laboratory Technicians work. The MLT program includes one or more courses in each of these areas.

The sections/departments in this area include:

- **Support Services** – This includes phlebotomy and processing. (Phlebotomy course)
 - **Phlebotomy** – This area is responsible for drawing blood from patients
 - **Processing** – Phlebotomists draw samples and send to this area for centrifugation and aliquoting.
- **Hematology** – This area determines number of cells (red blood cells, white blood cells, platelets) in the blood and look at changes in cells. (Basic Hematology and Advanced Hematology courses)
- **Coagulation** – Department where tests are performed to see how a patient's blood is clotting. Tests are performed to monitor treatment of blood clotting disorders. (Coagulation course). This department is sometimes within the hematology area, especially in smaller facilities.
- **Urinalysis** – Physical, chemical and microscopic examination of urine is performed in this department (v). As with the coagulation area, urinalysis is sometimes in hematology.
- **Chemistry** – The Chemistry department utilizes complex instruments to determine concentration of substances (glucose, cholesterol) in samples. (Clinical Chemistry course). This is typically the lab department with the highest volume of tests.
- **Immunology/Serology** – Many tests in this department are based on antigen/antibody reactions. Common tests include pregnancy, HIV, hepatitis, and infectious disease testing. (Basic Immunology Concepts course) This department is sometimes within the chemistry area, especially in smaller facilities.
- **Blood Bank/Transfusion Services/Immunohematology** – This area determines the patient's blood type (ABO and Rh) and what blood product will be compatible to transfuse the patient. (Blood Bank course)
- **Microbiology** – This department cultures and identifies microorganisms causing disease and determines antibiotics that will be effective against the organism. (Clinical Microbiology and Advanced Microbiology courses)

- **Molecular Diagnostics/Molecular Testing** – Area performing analysis nucleic acid (DNA/RNA) to detect and identify diseases or genetic disorders. Most often it is incorporated into other lab areas such as microbiology. (Introduction to Molecular Diagnostics course).
- **Referrals** – Department that prepares samples for shipment to other testing facilities.

Anatomic and Surgical Pathology Departments

The other main area of the lab is the anatomic and surgical pathology area, where they analyze the characteristics of cells and tissues.

The sections/departments in this area include:

- **Cytogenetics** – The department that examines chromosomes for evidence of genetic disease.
- **Cytopathology** – This area processes and stains cells in fluids and examines them.
- **Histology** – This department prepares tissues for exam by the pathologist.

Laboratory Personnel

Similar to nursing, there are several levels of education, training, and skills found in laboratory workers. The Clinical Laboratory Improvement Amendment of 1988¹ defines the qualifications of the personnel based on the job functions required of that position.

Clinical Pathology Personnel

- **Laboratory Director** – This is typically a pathologist (medical doctor trained in laboratory medicine)
- **Laboratory Manager** – An individual trained in lab science with additional healthcare administration or business/management degree or experience. The Lab Manager reports to the Lab Director.
- **Department Supervisor** – This person is responsible for the work performed in that respective department/lab area. They report to the Lab Manager. This person may be an MLT or MLS depending on the requirements of specific employers.
- **Laboratory Information System (LIS) Analyst** – This individual works with the computer systems of the laboratory and the hospital's electronic medical record.
- **Testing Personnel** – These are technicians or scientists performing testing.
 - **Medical Laboratory Technician (MLT)/Clinical Laboratory Technician (CLT)** – These technicians have an Associate's degree and clinical training. They may have completed a national certification exam to become certified, though certification is not always required.
 - **Medical Laboratory Scientist (MLS)/Clinical Laboratory Scientist (CLS)** – These personnel have a Baccalaureate degree and clinical training. They may have completed a national certification exam to become certified, though certification is not always required.
- **Laboratory Assistant** – This individual receives and processes specimens, answers the phone, and may perform phlebotomy. A high school diploma is required.
- **Phlebotomist** – These individuals draw blood samples and process samples. A high school diploma is required. These individuals may have on the job training or have taken a phlebotomy course.

Anatomic and Surgical Pathology Personnel

- **Pathology Assistant** – These individuals assist the pathologist with their daily duties. These duties include collecting and processing samples and preparing reports to describe their findings.
- **Histotechnician/Histotechnologist** – These individuals prepare and analyze biological tissues for microscopic examination.
- **Cytotechnologist** – These individuals process cellular samples obtained through pap smears, needle aspirations, and biopsies.

Certification Agencies

To become a certified laboratory professional, individuals must complete an approved course of study and pass a national certification examination. There are several types of training programs offered. They can be through colleges and universities, hospitals, technical schools, or private institutions. There is typically an academic component as well as a clinical component to the training programs. Certification agencies are independent institutions that provide examinations for the various levels of laboratory professionals. Examples include: [American Medical Technologists \(AMT\)](#) and the [American Society for Clinical Pathology \(ASCP\)](#). Each organization defines the requirements to be eligible to sit for their certification exams and the continuing education requirements necessary to maintain the certification. In our area, the ASCP Board of Certification exam is the most commonly obtained certification.

Relationship with Other Health Professionals

The clinical laboratory professional is a valued member of the healthcare team. They work with nurses, physicians, therapists, and other professionals to provide quality patient care. Many clinical decisions are based on the laboratory test results that MLTs provide.

Additional Resources

- "[Clinical Laboratory Technologists and Technicians](#)," U.S. Department of Labor, *Occupational Outlook Handbook*.
- "[Careers in Laboratory Science](#)," Coordinating Council on the Clinical Laboratory Workforce, *LaboratoryScienceCareers.com*.
- "[ASCP Board of Certification](#)," American Society for Clinical Pathology, *ASCP.org*.

References

¹[102 Stat. 2903, Public Law 100-578](#)

This page titled [5: Health Careers](#) is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by [Kari Christenson](#), [Western Technical College](#).

6: CLIA

Learning Objectives

Course Competency:

Evaluate laboratory compliance with CLIA regulations.

Learning Objectives:

- Describe the major components of CLIA
- Explain the relationship of one CLIA component to another
- Demonstrate understanding of regulatory compliance
- Explain the proficiency testing process

Clinical Laboratory Improvement Amendments (CLIA)¹

All clinical laboratories that perform tests on human specimens are regulated by both federal and state agencies. Congress passed the Clinical Laboratory Improvement Amendments (CLIA) in 1988, setting quality standards for all lab testing. CLIA ensures the accuracy, reliability, and timeliness of patient test results, regardless of where the test is performed. The CLIA regulations are based on the complexity of the test method. That means the more complicated the test, the more stringent the requirements. The type of CLIA certificate a lab must have depends on the type of testing the lab performs.

Under CLIA, a clinical laboratory is defined as a facility that performs testing on materials derived from the human body for the purpose of providing information for the diagnosis, prevention, or treatment of any disease or impairment of, or the assessment of health of human beings. These exams also include procedures to determine, measure, or otherwise describe the presence or absence of various substances or organisms in the body. If a facility performs tests for these purposes, regardless of whether there is reimbursement for such tests, it is considered a laboratory according to the definition by CLIA. Therefore, it must obtain and maintain CLIA certification.

Although CLIA is a federal program administered by Centers for Medicare & Medicaid Services (CMS), state agencies are responsible for overseeing labs and maintaining CLIA labs' certification records. State agencies process CLIA applications, renewals, updates, and requests for certificate copies. In addition, state agencies inspect and monitor clinical labs for compliance of federal CLIA regulations. They also investigate complaints to ensure testing is accurate, effective, and reliable.

CLIA Categorizations²

The FDA categorizes diagnostic tests by their complexity—from the least to the most complex: **waived tests**, **moderate complexity tests**, and **high complexity tests**. Classifications are based on the difficulty of the test procedures and the level of training required to accurately perform the tests. Standards for laboratory personnel differ for each category. The more complex the test, the more highly trained the testing personnel must be. The CLIA Program grants 5 types of lab certificates and each laboratory must obtain a certificate stating its classification.

Waived Tests

Diagnostic tests are categorized as waived based on the premise that they are simple to use, and there is little chance the test will provide wrong information or cause harm if it is done incorrectly. Tests that are cleared by the FDA for home or over-the-counter use are automatically assigned a waived categorization.

For a complete list, refer to the [CDC's Waived Test List](#).

Nonwaived Tests

Moderate and high complexity tests are also called nonwaived tests. The FDA determines the test's complexity by reviewing the package insert test instructions and using a criteria "scorecard" to categorize a test as moderate or high complexity. Each test is graded for level of complexity by assigning scores of 1, 2, or 3 for each of the seven criteria on the scorecard.

CLIA Categorization Criteria³

- Knowledge
- Training and experience
- Reagents and materials preparation

- Characteristics of operational steps
- Calibration, quality control, and proficiency testing materials
- Test system troubleshooting and equipment maintenance
- Interpretation and judgment

A score of 1 indicates the lowest level of complexity, and the score of 3 indicates the highest level. The 7 scores are added together and the tests with a score of 12 or less are categorized as moderate complexity, and those with a score above 12 are categorized as high complexity.

Additional information is available at the [FDA's CLIA Categorizations webpage](#).

Survey Process⁴

Labs are surveyed or inspected to ensure that CLIA regulations are met. Surveys may be scheduled in advanced or unannounced. The survey process is outcome-oriented, the intent of which is to focus the surveyor on the overall performance of the laboratory and the way it monitors itself, rather than on a methodical evaluation of each requirement. During the information gathering portion, surveyors review a variety of records including the lab's operating procedures, quality assurance records, proficiency testing reports, personnel records, and patient test results. They also conduct observations and interviews.

The surveyor analyzes all collected information to determine whether the laboratory has complied with applicable federal regulations. The surveyor reviews these findings in an exit conference and summarizes the survey findings in a written final report. Any deficiencies are discussed and addressed between the surveyor and the lab.

For additional information about the Wisconsin survey process, refer to the [WI Department of Health Services \(DHS\) Survey Guide: Clinical Laboratories, P-01227](#).

Proficiency Testing⁵

CLIA regulations require a laboratory to have quality control procedures to ensure accuracy and reliability of test results. Proficiency testing (PT) is the process used to evaluate a laboratory's performance and maintain quality control among laboratories.

Enrollment in an external proficiency testing program is required for laboratories performing moderate or high complexity tests. Only waived tests are exempt from proficiency testing regulations. If a laboratory performs only waived tests, it is not required to participate in a proficiency testing program. However, it must apply for and be given a certificate of waiver.

PT is a useful tool to:

- Verify testing accuracy and reliability
- Validate the entire testing process, including the testing personnel competency.

PT reports can alert laboratory staff and the laboratory director to testing phases (for example, pre- or post-analytical) that aren't performing as expected. Routinely reviewing PT reports can also indicate subtle shifts and trends that, over time, could affect patient results.

PT Process

Proficiency testing programs are available through the College of American Pathologists, the Centers for Disease Control and Prevention, and the health departments in some states (ex. Wisconsin State Lab of Hygiene). An HHS-approved external PT program sends 5 samples to the laboratory on a regular basis (usually 3 times each year) for testing. Each shipment of PT samples is called an "event."

Labs must test PT samples the same way they test patient specimens. This means:

- Testing the PT samples the same number of times as labs would test patient specimens.
- Testing at the same time as patient specimens.
- Testing by the same personnel that routinely test the patient specimens.
- Using the same test system, including analyzer and reagents, that you routinely use for the patient specimens. PT samples should also be rotated among the laboratory's testing personnel.

Laboratories must not communicate with other labs regarding PT values and the lab must not send the PT samples to another lab.

The laboratory must also document its process for all PT samples and this information must be kept for two years. When testing is complete, the laboratory reports its results to the PT program, which then grades the results against CLIA performance criteria and

sends the laboratory a score. All PT results must then be reviewed, signed, and dated by the lab. CMS and accreditation organizations routinely monitor each laboratory's score and overall performance.

Performance Criteria and PT Failure

CMS specifies acceptable performance criteria for each analyte. An 80% or better is acceptable (0 or 1 wrong out of the 5 samples), but failing even a single analyte requires the lab to investigate and correct the problem as well as document their actions. More than 1 incorrect result is a failure for that analyte in that event. CMS labels this "unsatisfactory" and requires corrective action.

If multiple PT events are failed, CMS labels this as "unsuccessful" and the lab would need to create a plan for correction of the problem and undergo ongoing monitoring. Repeated unsuccessful performance may result in the laboratory no longer being allowed to perform the failed testing. In addition, CMS may revoke the lab's certificate and stop Medicare/Medicaid payments.

Common causes of PT failure include missing or inadequate samples, test procedural errors, improperly calibrated instruments, incomplete forms, transcription errors, and failure to return samples by the deadline. All failures must be investigated to determine the type of error and correct the problem. Labs must also document all corrective action taken.

Competency and Performance Assessment⁶

The ability of each person to perform their duties should be assessed following their initial training and periodically thereafter. Retraining and reassessment of employee performance needs to be done when problems are identified with employee performance. The training and assessment program should be documented and specific for each job description. Activities requiring judgment or interpretive skills need to be included in the assessment.

Performance assessment can:

- Identify key training areas
- Identify processes that need improvement
- Provide supervisors and managers with data on employee performance
- Provide evidence to customers and management that the laboratory testing site assures quality with trained staff

Some elements of performance assessment include:

- Observing routine patient test performance, including sample handling, processing, and testing
- Monitoring recording and reporting of test results
- Reviewing intermediate test results or worksheets, QC records, proficiency testing results, and preventive maintenance records
- Observing performance of instrument maintenance and function checks
- Assessing test performance through testing previously analyzed samples, internal blind testing samples, or external proficiency testing samples
- Evaluating problem-solving skills

View [an example Performance Assessment in Appendix E \(pages 46-48\) of the CDC's document, "To Test Or Not To Test."](#)

References

¹ This section is minimally adapted from [Clinical Laboratory Improvement Amendments \(CLIA\)](#), Wisconsin Department of Health Services (DHS). Retrieved January 28, 2025. Used with permission from the State of Wisconsin.

² Adapted from [8.3 Regulation of Clinical Tests](#), Jack O'Grady, Austin Community College. CC BY.

³ [CLIA Categorizations](#), U.S. Food and Drug Administration. [Public Domain](#). Retrieved January 28, 2025.

⁴ This section is adapted from the [Survey Guide for Clinical Laboratories](#), Wisconsin Department of Health Services (DHS). Retrieved January 28, 2025. Used with permission from the State of Wisconsin.

⁵ [Proficiency Testing and PT Referral](#) (2024, June), Centers for Medicare & Medicaid Services. [Public Domain](#).

⁶ [To Test or Not to Test](#) (2024, June), U.S. Centers for Disease Control and Prevention, [Public Domain](#).

This page titled [6: CLIA](#) is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by [Kari Christenson, Western Technical College](#).

- [8.3: Regulation of Clinical Tests](#) by Jack O'Grady is licensed [CC BY 4.0](#).

Index

B

balance

2: Basic Lab Equipment

biological hazards

1: Lab Safety

C

centrifuge

2: Basic Lab Equipment

certification agencies

5: Health Careers

chain of infection

1: Lab Safety

chemical hazards

1: Lab Safety

CLIA regulations

6: CLIA

Clinical Laboratory Improvement

Amendments (CLIA)

6: CLIA

combining form

4: Medical Terminology

combining vowel

4: Medical Terminology

G

glassware

2: Basic Lab Equipment

H

Hazard Communication Standard

1: Lab Safety

hazards

1: Lab Safety

health career options

5: Health Careers

I

infection control

1: Lab Safety

L

lab equipment

2: Basic Lab Equipment

3: Microscope

laboratory compliance

6: CLIA

laboratory personnel

5: Health Careers

M

medical terminology

4: Medical Terminology

medical terms

4: Medical Terminology

micropipette

2: Basic Lab Equipment

microscope

3: Microscope

P

personal protective equipment

1: Lab Safety

physical hazards

1: Lab Safety

pipette

2: Basic Lab Equipment

prefix

4: Medical Terminology

S

safety

1: Lab Safety

standard precautions

1: Lab Safety

suffix

4: Medical Terminology

W

waived testing

6: CLIA

word root

4: Medical Terminology

Glossary

accuracy | True or target value determined by comparing results to accepted correct results or by comparing to results from another laboratory using a comparable method

analyte | A substance or constituent for which the laboratory conducts testing

balance | A laboratory instrument used to measure the mass of substances. Analytical balances provide high precision measurements, often to the nearest 0.0001 g.

beaker | A cylindrical container with a flat bottom, commonly used for mixing, heating, and holding liquids. Beakers are typically made of glass or plastic and come in various sizes.

binocular | Refers to instruments, such as microscopes, that are designed for use with both eyes simultaneously.

biological hazards | Microorganisms or biological materials that pose a threat to human health, including bacteria, viruses, fungi, and parasites.

Blood Borne Pathogen (BBP) | Microorganisms present in human blood that can cause disease in humans, including Hepatitis B, Hepatitis C, and HIV.

Borosilicate glass | A type of glass resistant to thermal shock and chemical corrosion, commonly used in laboratory equipment like test tubes and beakers.

Center for Medicare and Medicaid Services (CMS) | A U.S. government agency overseeing the administration of national health programs including Medicare, Medicaid, and the Affordable Care Act.

centrifuge | A device that uses rapid spinning to separate components of a mixture based on their density, commonly used in blood and urine analysis.

chemical hazards | Substances that pose a risk to health or the environment, including corrosive, flammable, toxic, or reactive chemicals.

chemical hygiene plan | A written program developed and implemented by a laboratory to ensure safe handling, storage, and disposal of hazardous chemicals.

Clinical Laboratory Improvement Act (CLIA) | U.S. federal regulations that establish quality standards for laboratory testing to ensure the accuracy, reliability, and timeliness of patient test results.

combining form | A word root combined with a combining vowel, forming a component of a medical term.

combining vowel | A word component, typically the letter "o," that helps with pronunciation. A combining vowel can connect a word root and a suffix or two word roots.

condenser | A component of a microscope that concentrates and directs light onto the specimen, enhancing contrast and resolution to reveal finer details of the microscopic specimen

course focus | A microscope adjustment used to bring objects into focus when using lower-power objective lenses.

critical measurement | A measurement that directly impacts the accuracy and reliability of test results, often requiring precise calibration and technique.

disposable pipette | A single-use instrument for transferring small volumes of liquids, typically made of plastic, to prevent cross-contamination.

engineering controls | Physical modifications to the workplace that reduce or eliminate exposure to hazards, such as fume hoods or safety shields.

fine focus | A microscope adjustment used to bring objects into sharp focus, particularly when using high-power objective lenses.

flask | A type of laboratory glassware with a narrow neck and wide body, used for mixing, heating, or storing liquids.

flint glass | A type of optical glass with a high refractive index, often used in lenses and prisms due to its dispersion properties.

Food and Drug Administration (FDA) | A U.S. government agency responsible for regulating food, drugs, medical devices, and other health-related products to ensure safety and efficacy

graduated cylinder | A tall, narrow container with marked increments, used to measure the volume of liquids accurately.

graduated pipette | A pipette marked with graduations along its length, allowing for the measurement and transfer of specific volumes of liquid.

Hazard Communication Standard | Occupational Safety and Health Administration (OSHA) regulation requiring employers to inform and train employees about hazardous chemicals in the workplace through labeling, safety data sheets, and training programs.

healthcare associated infection | Infections acquired by patients during the course of receiving treatment for other conditions within a healthcare setting

Hepatitis B virus | A virus that causes liver infection, transmitted through contact with infected blood or body fluids. Vaccination is available and recommended.

Hepatitis C virus | A bloodborne virus that can cause chronic liver disease and is primarily transmitted through blood-to-blood contact.

highly complex | Laboratory tests that require specialized knowledge, training, and equipment, often involving multiple steps and quality control measures.

Human Immunodeficiency Virus (HIV) | A virus that attacks the immune system, leading to acquired immunodeficiency syndrome (AIDS). Transmitted through contact with certain body fluids.

iris diaphragm | A component of a microscope that adjusts the amount of light passing through the specimen, enhancing contrast and resolution.

meniscus | The curved surface of a liquid in a container, such as a graduated cylinder, where measurements are read at the bottom of the curve.

micropipette | A precision instrument used to measure and transfer very small volumes of liquids, typically in microliters.

moderately complex | Laboratory tests that require some specialized knowledge and training but are less intricate than highly complex tests.

noncritical measurement | A measurement that does not significantly impact the accuracy or reliability of test results, often requiring less stringent controls.

objective lens | The lens in a microscope closest to the specimen, responsible for gathering light and resolving details of the sample.

Occupational Safety and Health Administration (OSHA) | A U.S. government agency that ensures safe and healthy working conditions by setting and enforcing standards and providing training and assistance.

ocular | Relating to the eyes; in microscopy, it often refers to the eyepiece through which the specimen is viewed.

other potentially infectious material (OPIM) | Any bodily fluids (other than blood) that may contain pathogens, including semen, vaginal secretions, cerebrospinal fluid, and materials contaminated with these fluids. Defined by OSHA to include materials that pose a risk of infection.

Personal Protective Equipment (PPE) | Specialized clothing or equipment worn by an employee for protection against a hazard. Examples of PPE are gloves, respirators, lab coats, and safety glasses.

physical hazards | Sources of potential harm due to environmental or mechanical conditions, such as noise, radiation, extreme temperatures, sharp instruments, or slips and falls in the laboratory.

pipette | A laboratory tool used to measure and transfer precise volumes of liquid. Pipettes come in various types, including manual, electronic, graduated, volumetric, and serological.

point of care testing (POCT) | Diagnostic testing performed at or near the site of patient care rather than in a central laboratory. Examples include glucose testing, pregnancy tests, and rapid strep tests.

precision | The degree to which further measurements or calculations show the same or similar results. This is also called reproducibility or repeatability.

prefix | A word component that appears at the beginning of the medical term and adds meaning to the word root. Prefixes often indicate number, position, direction, time, or negation.

proficiency testing (PT) | A method used to evaluate the accuracy and reliability of a laboratory's test results by comparing them to those from an external source or standard. Required under CLIA for many types of testing.

quartz | A hard, chemically resistant, and heat-tolerant mineral often used in laboratory instruments like cuvettes and specialized glassware due to its optical clarity and durability.

rheostat | An adjustable resistor used to control the intensity of electrical current in devices like microscope light sources or incubators.

Safety Data Sheet (SDS) | A document that provides detailed information on the properties, hazards, safe handling, and emergency measures related to a chemical substance. Formerly called Material Safety Data Sheet (MSDS).

serological pipette | A graduated pipette used to measure and transfer liquid volumes in a lab. These pipettes are typically calibrated to deliver a specific volume including the tip.

stage | The flat platform of a microscope on which slides are placed. The stage often includes clips or a mechanical stage to hold the slide in place during observation.

standard precautions | A set of infection control practices used to prevent transmission of diseases through contact with blood, body fluids, non-intact skin, and mucous membranes. These precautions include hand hygiene, use of PPE, and safe needle practices.

suffix | A word component at the end of the medical term that usually indicates a procedure, condition, or disease.

test tube | A cylindrical piece of glassware used for holding, mixing, or heating small quantities of liquids. Test tubes are commonly used for chemical reactions and biological samples.

transfer pipette | A simple pipette, usually plastic and disposable, designed for one-time use in transferring small, approximate volumes of liquid. Common in point-of-care and general lab use.

volumetric pipette | A precision pipette calibrated to deliver a single, specific volume of liquid. Used when high accuracy is required.

waived tests | Tests determined by the FDA to be simple with low risk of an incorrect result or with no reasonable risk of harm

word root | A word component that is the core of many medical terms and refers to the body part or body system to which the term is referring.

workplace controls | Procedures and practices implemented to reduce or eliminate exposure to hazards. Examples include handwashing protocols, proper disposal methods, and training in spill response.

Detailed Licensing

Overview

Title: [Basic Lab Skills \(Christenson\)](#)

Webpages: 18

Applicable Restrictions: Noncommercial

All licenses found:

- [CC BY-NC-SA 4.0](#): 83.3% (15 pages)
- [Undeclared](#): 16.7% (3 pages)

By Page

- [Basic Lab Skills \(Christenson\)](#) - [CC BY-NC-SA 4.0](#)
 - [Front Matter](#) - [CC BY-NC-SA 4.0](#)
 - [TitlePage](#) - [CC BY-NC-SA 4.0](#)
 - [InfoPage](#) - [Undeclared](#)
 - [Introduction](#) - [CC BY-NC-SA 4.0](#)
 - [Table of Contents](#) - [Undeclared](#)
 - [Acknowledgements](#) - [CC BY-NC-SA 4.0](#)
 - [Licensing](#) - [Undeclared](#)
 - [1: Lab Safety](#) - [CC BY-NC-SA 4.0](#)
 - [2: Basic Lab Equipment](#) - [CC BY-NC-SA 4.0](#)
 - [3: Microscope](#) - [CC BY-NC-SA 4.0](#)
 - [4: Medical Terminology](#) - [CC BY-NC-SA 4.0](#)
 - [5: Health Careers](#) - [CC BY-NC-SA 4.0](#)
 - [6: CLIA](#) - [CC BY-NC-SA 4.0](#)
 - [Back Matter](#) - [CC BY-NC-SA 4.0](#)
 - [Index](#) - [CC BY-NC-SA 4.0](#)
 - [Glossary](#) - [CC BY-NC-SA 4.0](#)
 - [Detailed Licensing](#) - [CC BY-NC-SA 4.0](#)